HierarchicalMamba: A Multiscale State Space Model With Dual Transitions for Hyperspectral Image Classification

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***Abstract*—Hyperspectral image (HSI) classification faces fundamental challenges: existing convolutional neural networks capture only local spatial patterns while neglecting spectral sequence dependencies, recurrent neural networks suffer from gradient vanishing and computational inefficiency, and Transformer architectures encounter quadratic computational complexity $O(L^2)$ limiting scalability for sequences with hundreds of spectral bands. This article proposes the HierarchicalMamba algorithm, which addresses these challenges through three core innovations leveraging state space models (SSMs) with linear complexity $O(L)$. First, we design a unified spatial-sequential modeling framework that encodes 2-D spatial coordinate information into 1-D sequence positional indices, enabling simultaneous modeling of spectral correlations and spatial neighborhood relationships within a single SSM framework while avoiding the complexity of traditional dual-stream architectures. Second, we develop a multiscale SSM decomposition theory through heterogeneous parallel branches and combinations of different activation functions, comprehensively capturing multigranularity features in hyperspectral data and introducing adaptive fusion weights to achieve optimal feature combination. Third, we innovatively propose a dual state transition (DST) mechanism that applies the state transition matrix twice within each computational cycle, significantly enhancing the modeling capability for higher-order state dependencies and complex nonlinear dynamics. Comprehensive experiments on five benchmark datasets including Indian Pines, Pavia University, Salinas Valley, Kennedy Space Center, and Houston2018 demonstrate that HierarchicalMamba achieves overall accuracies of 96.11%, 98.03%, 97.06%, 95.34%, and 82.35% respectively, significantly outperforming existing methods while maintaining linear computational complexity and featuring 3.04M parameters, training time of 5.87 s/epoch on Pavia University (approximately 53× faster than TRANSF_uniformer_tiny at 313.33 s/epoch), and inference time of 2.51 ms/image. This method provides a new theoretical framework and technical pathway for HSI classification. The source code for HierarchicalMamba will be made publicly available at: https://github.com/Jason20155/HierarchicalMamba**

***Index Terms*—Dual state transition, hierarchical feature learning, hyperspectral image (HSI) classification, spatial-sequential modeling, state space models (SSMs).**

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I. Introduction

HYPERSPECTRAL image (HSI) classification facilitates precise material identification through detailed spectral signatures across hundreds of contiguous wavelengths, advancing precision agriculture, environmental monitoring, geological exploration, and urban planning. However, the high-dimensional nature of hyperspectral data—characterized by rich spectral information yet limited spatial resolution—presents fundamental computational and methodological challenges requiring innovative architectural solutions.

Traditional machine learning approaches established foundational methodologies for HSI classification. Support Vector Machines dominated early developments due to their effectiveness in high-dimensional spaces, with advances in distance metric learning, dimensionality estimation, and sparse regularization [1], [2], [3]. Subspace-based methods including multinomial logistic regression [4] and non-negative tensor factorization [5] extended spectral-spatial modeling capabilities, while decision tree approaches [6], [7], [8] demonstrated computational efficiency. Architectural innovations in general computer vision have informed hyperspectral methodologies. Li et al. [9] developed propagation-based CNNs for stereo segmentation, Zhao et al. [10] applied generative adversarial networks with dilated convolutions to medical imaging, and Zhao and Yu [11] proposed heterogeneous network embedding for multi-view modeling. Multi-objective feature selection methods reduce dimensionality through preprocessing rather than end-to-end learning [12]. However, these methods treated pixels as independent samples, neglecting spatial context and spectral sequence correlations, thereby constraining performance in complex scenarios.

CNN-based methods have advanced HSI classification through hierarchical feature learning. 2-D architectures by Benmouna et al. [13] and 3-D designs by Chen et al. [14] and Chen et al. [15] enabled joint spectral-spatial modeling. Lightweight frameworks by Fang et al. [16] and Ge et al. [17], domain-specific applications by Heo et al. [18] and Said et al. [19], and multi-object fusion methods by Hu et al. [20] and Qu et al. [21] expanded CNN capabilities. However, local

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receptive fields constrain long-range dependency modeling and substantial labeled data remain necessary.

Recurrent neural network (RNN) methods explored sequential modeling of spectral information. Research by Dasu et al. [22], Liu et al. [23], and Venkatesan and Prabu [24] investigated treating spectral bands as sequential data, yet encountered fundamental limitations including gradient vanishing, computational inefficiency, and parallelization constraints when processing high-dimensional hyperspectral sequences.

Attention mechanisms enhanced CNNs through selective feature emphasis. Spatial attention methods by Abdulsamad et al. [25], Li and Wang [26], and Li et al. [27] identified relevant spatial regions, while spectral attention by Fang et al. [28] and Praveen and Menon [29] selected discriminative bands. Liu et al. [30], [31], [32] developed central attention networks, multiarea target attention for multiscale objects, and cluster attention for semantic segmentation. Multi-dimensional fusion approaches by Ashraf et al. [33], Huang et al. [34], and Pan et al. [35] integrated multiscale information. Center attention designs by Feng et al. [36], pure attention architectures by Liao et al. [37], cross-attention mechanisms by Liu et al. [38], and residual attention networks by Wang et al. [39] and Zhang et al. [40] demonstrated architectural diversity. MSDANet by Liao and Wang [41] combined parallel multi-branch structures with dual-channel-spatial attention, employing depth-wise separable convolutions for computational efficiency. However, these CNN-based attention approaches remain constrained by the inherent locality of convolutional operations, limiting their capacity to capture global spectral-spatial dependencies across the entire hyperspectral cube.

Transformer architectures displayed sequence modeling potential for HSI classification. Positional encoding innovations by Ahmad et al. [42] and neighborhood attention by Arshad et al. [43] addressed hyperspectral characteristics, while multi-branch designs by Bai et al. [44] and foundational frameworks by He et al. [45] established spatial-spectral joint modeling. Semi-supervised approaches by He et al. [46] addressed label scarcity, while hybrid architectures by Li et al. [47] and Rehman et al. [48] combined methodological advantages. Window attention by Liu et al. [49], improved designs by Qing et al. [50], selective fusion by Xu et al. [51] and Zhang et al. [52], multisource fusion by Zhao et al. [53], and super-resolution applications by Zheng et al. [54] expanded technical diversity. Recent extensions include transductive few-shot learning by Xi et al. [55] for extreme low-sample scenarios and mixed CNN-Transformer with graph contrastive learning by Xi et al. [56] for Mars hyperspectral classification. However, these approaches encounter fundamental limitations: quadratic computational complexity $O(L^2)$ and dependence on large-scale pre-training data constrained scalability for high-dimensional hyperspectral sequences, while few-shot approaches require episodic meta-learning protocols and hybrid methods necessitate explicit dual-stream fusion mechanisms. Foundation models require extensive pretraining, contrasting with supervised HSI classification using limited labeled samples [57].

The emergence of Mamba and state space models (SSMs) in recent years has provided new technological paradigms for HSI classification, demonstrating advantages in maintaining sequence modeling capabilities while achieving linear computational complexity. Wavelet integration approaches such as WaveMamba by Ahmad et al. [58] integrated wavelet transformation with spatial-spectral Mamba architectures, grouping mechanisms such as IGroupSS-Mamba developed by He et al. [59] achieved efficient modeling through interval group partitioning, and foundational architectures such as SS-Mamba proposed by Huang et al. [60] established theoretical foundations in this field. Semantic tokenization approaches such as the STMamba model by Ming et al. [61] focused on semantic information representation, local enhancement methods such as LE-Mamba by Wang et al. [62] were optimized for noncausal feature extraction, and multiscale designs such as the Mamba-in-Mamba architecture by Zhou et al. [63] innovated feature aggregation mechanisms. Collaborative representation approaches such as MCIFNet by Zhu et al. [64] demonstrated the potential for fusion with implicit neural representations. Recent SSM-based architectures have explored specialized designs for hyperspectral classification, yet reveal persistent architectural limitations. DualPathmamba by Liao and Wang [65] introduced a dual-path architecture simultaneously processing spectral sequences through SSM operations and spatial features through convolutional operations, though the architectural separation necessitates explicit fusion mechanisms that complicate joint optimization of spatial-spectral dependencies. HyperspectralMamba proposed by Liao and Wang [66] integrated state space modeling with band-adaptive feature recalibration through a dual-stream design, yet maintains separation between global SSM processing and local convolutional extraction, requiring residual connections for feature integration rather than unified modeling within the state space framework. Although SSMs have obvious advantages in computational efficiency, existing methods mainly focused on single-dimensional sequence modeling, lacked effective mechanisms for unified modeling of spatial and spectral information, and exhibit certain limitations in multiscale feature fusion and complex state transition modeling, limiting the full utilization of the dual characteristics of spatial neighborhood relationships and spectral sequence dependencies in hyperspectral data.

To systematically clarify the motivation for adopting and advancing SSM architectures, Table I provides a comparative analysis of computational complexity and fundamental limitations across different methodological paradigms.

This comparison establishes SSMs as an optimal foundation with linear complexity and long-range modeling capabilities. However, systematic examination reveals three fundamental constraints that limit existing approaches and necessitate theoretical innovations.

First, spatial-spectral processing remains architecturally separated across most methodologies. CNN-based approaches inherently process spectral dimensions as independent channels, neglecting sequential dependencies among bands.

Attention mechanisms typically employ separate modules for spatial and spectral processing, necessitating explicit

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TABLE I
COMPARATIVE ANALYSIS OF ARCHITECTURAL PARADIGMS

Method	Complexity	Limitations	Proposed Approach
CNN	$O(K^2 \cdot H \cdot W \cdot C \cdot d)$	Local fields; independent channels	Long-range sequential modeling
Dual-branch	$2 \times O(L \cdot d)$	Separated pathways; redundancy	Unified encoding
Transformer	$O(L^2 \cdot d)$	Quadratic cost; overfitting	Linear cost; sample-efficient
SSM	$O(L \cdot d^2)$	Single-scale; 1st-order	Multi-scale; higher-order

fusion strategies. Recent SSM-based methods predominantly adopt dual-stream or dual-path architectures, processing spatial and spectral information through independent computational pathways before integration. This architectural separation introduces design complexity and computational redundancy while constraining joint optimization of spatial-spectral relationships.

Second, multiscale feature extraction relies predominantly on external mechanisms rather than intrinsic model formulations. CNN methods achieve multiscale processing through convolutional pyramids with varying kernel sizes. Attention architectures employ multi-head configurations or hierarchical structures implemented outside core attention computations. SSM-based approaches utilize auxiliary techniques including wavelet decomposition, spatial interval partitioning, or nested architectural designs—mechanisms operating externally to the state space formulation. These external strategies require additional parameters and architectural components without fundamentally extending the representational capacity of underlying models.

Third, modeling of long-range dependencies encounters inherent limitations across different architectures. RNN-based methods experience gradient degradation when processing sequences of hundreds of spectral bands. Transformer architectures provide global receptive fields but incur quadratic computational complexity $O(L^2)$, constraining scalability for high-dimensional hyperspectral data. Existing SSM methods, while achieving linear complexity $O(L)$, employ standard first-order state transitions that restrict capacity for modeling higher-order dependencies spanning noncontiguous spectral regions. Approximating such dependencies necessitates stacking multiple layers, increasing architectural depth and parameter count without advancing the theoretical framework.

These prevalent constraints—architectural separation of spatial-spectral processing, reliance on external multiscale mechanisms, and limitations in higher-order dependency modeling—indicate the necessity for fundamental theoretical innovations rather than incremental architectural refinements.

These constraints reveal a fundamental trade-off between feature expressiveness and architectural efficiency. Existing methods address limitations through separate mechanisms: dual-stream architectures for spatial-spectral processing [49], [50], external modules for multiscale features [41], [42], [46], [48], and layer stacking for dependency modeling [41], [42], [43], [44], [45], [46], [47]. Each solution introduces additional complexity and parameter overhead.

Our approach employs synergistic integration where three innovations form hierarchical dependencies. Unified spatial-sequential encoding creates a singular representation preserving both spatial proximity and spectral correlations, eliminating dual-stream requirements. SSM-internal multiscale decomposition operates on this unified sequence through heterogeneous parallel branches with distinct nonlinear characteristics, embedding multi-granularity extraction directly within the state space formulation rather than through external mechanisms. The dual state transition (DST) mechanism enables higher-order state dynamics across multiple scales through within-cycle dual matrix application, modeling complex spectral dependencies without network deepening.

This hierarchical structure distinguishes our design from independent component combination: unified encoding enables internal multiscale decomposition, which enables effective dual-transition modeling. The integrated architecture maintains linear computational complexity while achieving comprehensive multiscale and higher-order feature representation within a single-stream framework.

Based on the identified limitations and integrated design rationale, this article makes three principal contributions to HSI classification.

Contribution 1 (Unified Spatial-Sequential Modeling Framework): We develop a coordinate-to-index encoding mechanism that transforms 2-D spatial positions into 1-D sequence indices, enabling simultaneous modeling of spectral correlations and spatial neighborhood relationships within a single SSM framework. This approach eliminates architectural separation inherent in dual-stream designs, avoiding complexity and computational redundancy while maintaining linear computational complexity.

Contribution 2 (Multiscale SSM Decomposition Theory): We establish a theoretical framework embedding multi-granularity feature extraction within the state space formulation—rather than through external preprocessing or auxiliary modules—via three heterogeneous parallel branches: Gaussian error linear unit (GELU)-activated branches 1 and 3 model smooth spectral gradients, while rectified linear unit (ReLU)-activated branch 2 extracts sparse discriminative features. These complementary representations are adaptively fused through learnable Softmax-normalized weights tailored to dataset-specific spectral characteristics.

Contribution 3 (DST Mechanism): We introduce a within-cycle DST mechanism that applies the state transition matrix twice in each computational cycle, enabling second-order state dynamics without increasing network depth. This mechanism fundamentally enhances SSM expressiveness beyond standard first-order formulations, improving modeling capability for complex nonlinear dynamics and higher-order spectral dependencies. This dual-iteration structure-applying the state transition matrix sequentially within each cycle-extends first-order Markovian transitions to capture multistep state

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TABLE II
MATHEMATICAL NOTATION AND DIMENSIONS

Symbol	Description	Dimension	Symbol	Description	Dimension
$\mathbf{X}$	Input HSI	$B \times C \times H \times W$	N	Number of hierarchical blocks	scalar
B	Batch size	scalar	$\mathbf{W}_1, \mathbf{W}_2, \mathbf{W}_3$	Multi-scale projection matrices	$D \times D$
C	Number of spectral bands	scalar	$\mathbf{h}$	Temporally modulated features	$B \times L \times D$
H, W	Spatial height and width	scalar	$\mathbf{g}_{state}$	Adaptive state gate	$B \times L \times d_s$
L	Sequence length ($L = H \times W$)	scalar	$\mathbf{h}_{state}$	First state transition output	$B \times L \times d_s$
D	Model feature dimension	scalar	$\mathbf{h}_{gated}$	Gated state features	$B \times L \times d_s$
d_s	State space dimension	scalar	$\mathbf{h}_{proj}$	Back-projected features	$B \times L \times D$
$\mathbf{A}$	State transition matrix	$D \times d_s$	$\mathbf{h}_{state2}$	Second state transition output	$B \times L \times d_s$
$\mathbf{B}$	Input projection matrix	$D \times d_s$	$\mathbf{h}_{out}$	SSM output features	$B \times L \times D$
$\mathbf{C}$	Output projection matrix	$D \times d_s$	$\mathbf{h}_{direct}$	Direct path features	$B \times L \times D$
$\mathbf{D}$	Direct feedthrough vector	D	$\boldsymbol{\alpha}$	Learnable fusion parameters	$\mathbb{R}^3$
$\boldsymbol{\delta}$	Temporal step size parameter	D	$\boldsymbol{\omega}$	Normalized fusion weights	$\mathbb{R}^3$
$\mathbf{W}_g$	Gate projection matrix	$D \times d_s$			

Note: Bold symbols (**A, B, C, D**) denote matrices/vectors; plain italics (*B, C, D, L*) denote scalars.

evolution, analogous to higher-order numerical integration in dynamical systems.

These contributions synergistically address fundamental constraints in existing methods, providing a unified theoretical framework that maintains computational efficiency while achieving comprehensive feature representation for HSI classification.

The remainder of this article is organized as follows. Section II presents the methodology, Section III provides experimental validation on five benchmark datasets, and Section IV concludes with contributions and future directions.

II. METHOD

The HierarchicalMamba algorithm represents a deep learning architecture based on SSMs, specifically designed for HSI classification tasks. As illustrated in Fig. 1, the core theoretical foundation of this algorithm lies in treating the spectral dimensions of HSIs as sequential data with temporal characteristics, leveraging the long-range dependency modeling capabilities of SSMs to capture complex correlations among spectral bands. Critically, the introduced DST mechanism performs dual application of the state transition matrix per cycle, enabling higher-order dependency modeling beyond conventional single-transition architectures.

Traditional CNNs often treat spectral dimensions simply as channel information when processing hyperspectral data, neglecting the sequential characteristics and correlations among spectral bands. While RNNs can handle sequential information, they suffer from gradient vanishing and computational efficiency issues when modeling long-range dependencies. HierarchicalMamba addresses these limitations by introducing SSMs, which maintain sequential modeling capabilities while offering linear computational complexity advantages.

Additionally, the algorithm employs an innovative spatial-sequence transformation mechanism that encodes 2-D spatial information into the positional indices of 1-D sequences. This enables a unified SSM framework to simultaneously model spectral correlations and spatial neighborhood relationships, avoiding the complexity of designing dual-stream networks required by traditional approaches.

A. Detailed Mathematical Derivation and Theoretical Analysis

1) Input Processing and Spatial-Sequence Transformation: To facilitate understanding of the mathematical formulations, Table II summarizes all notation and dimensions used throughout this section.

Given a HSI $\mathbf{X}$, where notation is defined in Table II. Raw hyperspectral data are characterized by high-dimensional spectral features but relatively low spatial resolution, necessitating appropriate preprocessing to accommodate subsequent sequential modeling.

The algorithm first employs progressive convolutional projection to map high-dimensional spectral features into a unified model space

$$\mathbf{X}_{\text{proj}} = \text{Progressive-Conv}(\mathbf{X}). \tag{1}$$

This progressive projection employs a bottleneck architecture design principle (dimension reduction followed by expansion through sequential 1×1 and 3×3 convolutions) to efficiently extract multiscale spectral-spatial features. The three-stage process begins with spectral dimensionality reduction through 1×1 convolution, followed by spatial context aggregation via 3×3 convolution that captures local neighborhood relationships, and concludes with feature space alignment to the model's standard dimension. This progressive approach balances computational efficiency with feature expressiveness through initial compressing spectral redundancy, then enriching spatial context, and finally standardizing the representation for subsequent processing. Each convolutional layer is equipped with batch normalization and ReLU activation functions to ensure feature distribution stability and nonlinear expressiveness.

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To enable sequence modeling within the SSM framework, we transform the 2-D feature map into a 1-D sequence via row-major ordering. The spatial coordinates (i, j) are mapped to sequence indices via

$$\text{seq_idx}(i, j) = i \times W + j, \quad 0 \le i < H, \quad 0 \le j < W \tag{2}$$

yielding $\mathbf{x}_{\text{seq}} \in \mathbb{R}^{B\times L\times D}$ where $L = H \times W$ represents the sequence length. This encoding preserves spatial locality: horizontally adjacent pixels (i, j) and $(i, j+1)$ become consecutive sequence elements at positions k and $k + 1$ while vertically adjacent pixels (i, j) and $(i + 1, j)$ are separated by exactly W positions. This row-major structure enables the SSM to capture both local and regional spatial dependencies through its sequential modeling mechanism without requiring additional attention modules. Critically, this serialization maintains complete information preservation: sequence position encodes spatial coordinates while the D-dimensional feature vector at each position retains full spectral signatures across bands. State transitions $\mathbf{h}(t) = f(\mathbf{h}(t-1), \mathbf{x}(t))$ thus simultaneously capture spatial context evolution (via sequential ordering) and spectral correlations (via feature vectors), achieving unified spatial-spectral modeling that eliminates dual-stream architectural requirements.

2) Core Mechanisms of Hierarchical Mamba Block: Higher-order state dependencies extend beyond first-order Markovian transitions $h(t) = f(h(t-1), x(t))$ by incorporating multistep history: $h(t) = g(h(t-1), h(t-2), x(t))$. For hyperspectral classification, this enables modeling spectral patterns dependent on both immediate and noncontiguous correlations within sequence representations.

Higher-order state dependencies refer to the model's capacity to capture relationships where state evolution incorporates multistep historical information, extending beyond first-order Markovian dynamics where current state $h(t)$ depends solely on immediate previous state $h(t-1)$ and current input $x(t)$. In HSI classification, this enables modeling complex spectral patterns that depend on both immediate spectral neighbors and longer-range correlations within the sequence representation.

The HierarchicalMambaBlock serves as the core component of the entire algorithm, with its design based on theoretical frameworks of multiscale feature decomposition and state space modeling. This module receives input features $\mathbf{x} \in \mathbb{R}^{B\times L\times d_{\text{model}}}$ and outputs enhanced feature representations through complex multi-branch processing and state evolution.

a) Multiscale feature projection: The heterogeneous design decomposes spectral features through functionally distinct activations. GELU provides smooth differentiable properties for continuous spectral transitions, while ReLU imposes hard thresholding inducing sparsity for discriminative feature extraction. This duality captures both gradual variations and discrete boundaries inherent in hyperspectral data.

The algorithm employs three parallel feature projection branches to extract feature representations at different scales, each possessing distinct nonlinear characteristics

$$\mathbf{f}_1 = \text{GELU}\left(\text{LN}\left(\text{Linear}\left(\mathbf{x}\right)\right)\right) \tag{3}$$

$$\mathbf{f}_2 = \text{ReLU}\left(\text{LN}\left(\text{Linear}\left(\mathbf{x}\right)\right)\right) \tag{4}$$

where $\text{Linear}(\cdot)$ denotes the linear transformation $\mathbf{Wx} + \mathbf{b}$, with parameter matrix $\mathbf{W}$ and bias vector $\mathbf{b}$ learned independently for each branch. $\text{LN}(\cdot)$ represents layer normalization, which stabilizes the training process and accelerates convergence.

The theoretical foundation of this heterogeneous design stems from multiscale analysis theory in signal processing. The GELU activation function, defined as $\text{GELU}(x) = x \cdot \Phi(x)$ where $\Phi(x)$ is the cumulative distribution function of the standard Gaussian distribution, possesses smooth nonlinear characteristics and approximate gating effects, making it particularly suitable for processing gradual features and subtle variations among spectral bands in hyperspectral data. The ReLU activation function $\text{ReLU}(x) = \max(0, x)$ exhibits hard threshold characteristics

$$\mathbf{f}_3 = \text{GELU}\left(\text{LN}\left(\text{Linear}\left(\mathbf{x}\right)\right)\right) \tag{5}$$

introducing sparsity and extracting more robust discriminative features. By combining these two activation functions with different characteristics, the algorithm can simultaneously capture both continuous and discrete features in spectral data.

b) Enhanced SSM computation: For each feature branch $\mathbf{f}_i$, the algorithm executes a series of complex state space transformations. This process is based on the discretization theory of continuous-time SSMs, learning state transition matrices and input–output mappings to model the dynamic evolution of sequences.

1) *Adaptive State Gating Mechanism:* The state gating mechanism learns an adaptive mask for selective transmission of state information

$$\mathbf{g}_{\text{state}} = \sigma\left(\text{Linear}_{d_{\text{state}}}\left(\mathbf{f}_i\right)\right) \tag{6}$$

where d_{state} is the dimension of the state space, $\sigma(\cdot)$ is the sigmoid activation function with output range $(0, 1)$. The linear transformation $\text{Linear}_{d_{\text{state}}}(\cdot)$ maps d_{model}-dimensional features to d_{state}-dimensional state space. The theoretical significance of this gating mechanism lies in achieving selective transmission of state information, similar to the forget gate mechanism in LSTMs but more simplified and efficient.

2) *Temporal Step Modulation Mechanism:* Temporal step modulation is a key component of SSMs, controlling the speed and magnitude of state evolution

$$\mathbf{h} = \mathbf{f}_i \odot \exp\left(\boldsymbol{\delta}\right) \tag{7}$$

where $\boldsymbol{\delta} \in \mathbb{R}^{d_{\text{model}}}$ is a learnable temporal step parameter vector, and $\odot$ denotes element-wise multiplication (Hadamard product, multiplying corresponding elements). The exponential function $\exp(\cdot)$ ensures that temporal steps remain positive, conforming to the nonnegativity constraint of temporal steps in physical systems. The learnability enables the model to learn different evolution speeds for different feature dimensions, reflecting the physical intuition that different spectral bands in HSIs may possess different temporal scale characteristics.

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3) *DST Process:* Traditional SSMs typically perform only one state transition, while HierarchicalMamba introduces a DST mechanism to enhance the model's expressiveness:

First State Projection:

$$\mathbf{h}_{\text{state}} = \mathbf{h}\mathbf{A} \tag{8}$$

where $\mathbf{A}$ is the first state transition matrix, mapping the input feature space $\mathbb{R}^D$ to the state space $\mathbb{R}^{d_s}$ In classical state space theory, this corresponds to the discretization process of continuous-time systems.

Gated State Update:

$$\mathbf{h}_{\text{gated}} = \mathbf{h}_{\text{state}} \odot \mathbf{g}_{\text{state}}. \tag{9}$$

The gating mechanism is applied through element-wise multiplication, achieving selective transmission of state information. This step introduces content-dependent nonlinear control, enabling the model to dynamically adjust state update strategies based on input content. *State Back-Projection:*

$$\mathbf{h}_{\text{proj}} = \mathbf{h}_{\text{gated}}\mathbf{B}^{\text{T}}. \tag{10}$$

Matrix $\mathbf{B}$ re-projects the gated state information back to the feature space. This step corresponds to the state-to-output mapping process in SSMs.

Second State Transition:

$$\mathbf{h}_{state2} = \mathbf{h}_{\text{proj}}\mathbf{A}. \tag{11}$$

The state transition matrix $\mathbf{A}$ is applied again, achieving further evolution of the state. The theoretical significance of this DST design lies in enhancing the model's expressiveness, enabling it to capture more complex dynamic patterns. From a dynamical systems perspective, this corresponds to executing dual iterations within the system dynamics, capable of modeling higher-order state dependencies.

Final Output Projection:

$$\mathbf{h}_{\text{out}} = \mathbf{h}_{state2}\mathbf{C}^{\text{T}}. \tag{12}$$

Matrix $\mathbf{C}$ is responsible for converting final state information to output features. In state space theory, this corresponds to the role of the observation matrix.

Direct Connection Path:

$$\mathbf{h}_{\text{direct}} = \mathbf{f}_i \odot \mathbf{D}. \tag{13}$$

Parameter vector $\mathbf{D}$ provides direct input-to-output connections, similar to skip connections in ResNet. This design is based on the theoretical understanding that certain features may not require complex state evolution processes and are more effective when transmitted directly.

Final SSM Output:

$$\text{SSM}(\mathbf{f}_i) = \mathbf{h}_{\text{out}} + \mathbf{h}_{\text{direct}}. \tag{14}$$

The final output is a weighted combination of the state evolution path and the direct path, ensuring that the model can capture complex sequential dependencies while maintaining direct transmission of input information.

c) Adaptive multiscale fusion: Multiscale feature fusion is an important component of the algorithm, with its theoretical foundation rooted in optimal information integration principles from information theory. Traditional feature fusion methods typically employ fixed weights or simple averaging, lacking adaptability to different tasks and data characteristics. *Learnable Weight Mechanism:* The algorithm introduces an adaptive weight learning mechanism

$$\boldsymbol{\omega} = \text{Softmax}(\boldsymbol{\alpha}) \tag{15}$$

where $\boldsymbol{\alpha} = [\alpha_1, \alpha_2, \alpha_3]^{\text{T}}$ is a learnable weight parameter vector, and the Softmax function ensures non-negativity and normalization constraints $\sum_{i=1}^{3} \omega_i = 1$. Unlike fixed averaging, this mechanism automatically adjusts branch contributions during training, enabling discovery of dataset-specific optimal combinations. Datasets with distinct spectral characteristics may emphasize different branches based on task requirements.

Weighted Feature Combination:

$$\mathbf{f}_{\text{weighted},i} = \text{SSM}(\mathbf{f}_i) \cdot \omega_i, \quad i = 1, 2, 3. \tag{16}$$

Each feature branch's contribution is modulated by its corresponding weight. From an information-theoretic perspective, this corresponds to weighted combination of information from different sources based on information content, with weights ω_i reflecting the importance of the i*i*-th branch for final decision-making.

High-Dimensional Feature Space Construction: To preserve the independence of different scale features while achieving effective fusion, the algorithm first constructs a high-dimensional joint feature space

$$\mathbf{f}_{\text{combined}} = \text{Concat}\left(\left[\mathbf{f}_{\text{weighted},1}, \mathbf{f}_{\text{weighted},2}, \mathbf{f}_{\text{weighted},3}\right]\right). \tag{17}$$

The concatenation operation combines outputs from the three branches along the feature dimension, creating a $3d_{\text{model}}$-dimensional high-dimensional feature representation.

Nonlinear Feature Fusion: Intelligent feature compression is achieved through a two-layer feedforward network

$$\mathbf{f}_{\text{fused}} = \text{LN}\big(\text{Linear}_{d_{\text{model}}}\big(\text{GELU}\big(\text{LN}\big(\text{Linear}_{2d_{\text{model}}}\big(\mathbf{f}_{\text{combined}}\big)\big)\big)\big)\big). \tag{18}$$

This employs a standard expansion-compression architectural design where the first layer expands feature dimensions from $3d_{\text{model}}$ to $2d_{\text{model}}$, introducing nonlinear transformations through GELU activation; the second layer compresses features back to the standard dimension d_{model}. This design can learn complex feature interaction patterns and nonlinear combination relationships.

d) Attention enhancement mechanism: To further optimize the fused feature representation, the algorithm introduces an attention mechanism

$$\mathbf{A}_{\text{att}} = \sigma\left(\text{Linear}_{d_{\text{model}}}\left(\text{GELU}\left(\text{Linear}_{d_{\text{model}}/4}\left(\mathbf{f}_{\text{fused}}\right)\right)\right)\right). \tag{19}$$

This is a two-layer attention network employing a compression-expansion architectural design. The first layer compresses feature dimensions to $d_{\text{model}}/4$, forcing the network to learn compact feature representations; the second

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layer restores to original dimensions and generates attention weights through the sigmoid function. This design is based on information bottleneck theory, where dimensional compression can extract the most important discriminative information.

Feature Enhancement:

$$\mathbf{f}_{\text{enhanced}} = \mathbf{f}_{\text{fused}} \odot \mathbf{A}_{\text{att}}. \tag{20}$$

Attention weights modulate fused features through element-wise multiplication, achieving selective feature enhancement, highlighting discriminative features while suppressing redundant information.

e) Two-level residual connection design: The algorithm employs a two-level residual connection mechanism to ensure training stability and information propagation efficiency:

Intrablock Residual:

$$\text{HierarchicalMambaBlock}(\mathbf{x}) = \mathbf{f}_{\text{enhanced}} + \mathbf{x}. \tag{21}$$

This is the residual connection within each HierarchicalMambaBlock, ensuring that original input information can be directly transmitted to the output.

Interlayer Residual Connection:

$$\mathbf{x}^{(l)} = \mathbf{x}^{(l-1)} + \text{HierarchicalMambaBlock}\left(\text{LN}\left(\mathbf{x}^{(l-1)}\right)\right). \tag{22}$$

This is the residual connection between different HierarchicalMambaBlocks, employing a Pre-layer normalization (LN) structure (normalization before processing), which offers better gradient propagation characteristics and training stability.

Expanding the complete computation yields

$$\mathbf{x}^{(l)} = \mathbf{x}^{(l-1)} + \mathbf{f}_{\text{enhanced}} + \text{LN}\left(\mathbf{x}^{(l-1)}\right). \tag{23}$$

This design creates three independent information propagation paths: direct skip path $\mathbf{x}^{(l-1)}$, normalization path $\text{LN}(\mathbf{x}^{(l-1)})$, and complete processing path $\mathbf{f}_{\text{enhanced}}$, greatly enriching the network's expressiveness and optimization landscape.

B. Hierarchical Feature Learning

Through cascading L HierarchicalMambaBlocks, the algorithm achieves progressive learning from low-level spectral features to high-level semantic features

$$\mathbf{x}^{(l)} = \mathbf{x}^{(l-1)} + \text{HierarchicalMambaBlock}\left(\text{LN}\left(\mathbf{x}^{(l-1)}\right)\right), \quad l = 1, 2, \ldots, L \tag{24}$$

where $\mathbf{x}^{(0)} = \mathbf{X}_{\text{seq}}$ is the initial sequence input. This recursive structure ensures progressive refinement of feature representations, with each layer learning more abstract and discriminative feature representations based on the previous layer.

Due to the presence of residual connections, the final feature representation can be expressed as

$$\mathbf{x}^{(L)} = \mathbf{x}^{(0)} + \sum_{i=1}^{L} \text{HierarchicalMambaBlock}\left(\text{LN}\left(\mathbf{x}^{(i-1)}\right)\right). \tag{25}$$

This indicates that the final features are the accumulation of initial input and processing results from all layers, similar to additive models in ensemble learning, effectively integrating feature information from different levels.

C. Output Reconstruction and Classification

After hierarchical processing, the algorithm reconstructs sequence features into spatial form and performs classification prediction.

Spatial Reconstruction:

$$\mathbf{X}_{\text{final}} = \text{Permute}\left(\text{Reshape}\left(\mathbf{x}^{(L)}\right)\right). \tag{26}$$

This operation is the inverse of the spatial-sequence transformation in the forward process, remapping sequence positional information back to 2-D spatial coordinates.

Final Classification:

$$\mathbf{y} = \text{Classifier}\left(\text{GlobalAvgPool}\left(\text{Conv}\left(\mathbf{X}_{\text{final}}\right)\right)\right) \tag{27}$$

where $\text{Conv}(\cdot)$ is the final convolutional adjustment layer, $\text{GlobalAvgPool}(\cdot)$ is the global average pooling operation, and N_{classes} is the number of classification categories. Global average pooling aggregates spatial dimensional information into a single feature vector, based on spatial invariance assumptions, and offers better generalization performance.

D. Computational Complexity Analysis

To rigorously substantiate the linear scaling properties of HierarchicalMamba relative to standard Transformer architectures, this section provides per-layer complexity derivations with theoretical justifications.

1) Per-Layer Complexity Derivation: Input Projection Layer: The progressive three-stage bottleneck architecture [see (1)] performs spectral compression followed by spatial context aggregation

$$\text{FLOPs}_{\text{input}} = \frac{CDL}{4} + \frac{45D^2L}{8} = O\left(L\left(C + D^2\right)\right) \tag{28}$$

$$\text{Params}_{\text{input}} = \frac{CD + 45D^2}{8}. \tag{29}$$

HierarchicalMambaBlock Core Components:

a) Multiscale feature projection: Three parallel branches [see (3)–(5)] employ heterogeneous activations for multi-granularity feature extraction

$$\text{FLOPs}_{\text{multiscale}} = 3LD^2, \quad \text{Params}_{\text{multiscale}} = 3D^2 + 6D \tag{30}$$

b) Enhanced SSM—core innovation: The SSM component [see (6)–(14)] performs selective state evolution. The DST mechanism [see (8)–(12)] applies state matrix $\mathbf{A}$ twice

$$\mathbf{h}_{\text{state}} = \mathbf{hA} \xrightarrow{\text{gate}} \mathbf{h}_{\text{gated}} \xrightarrow{\mathbf{B}^{\text{T}}} \mathbf{h}_{\text{proj}} \xrightarrow{\mathbf{A}} \mathbf{h}_{state2} \xrightarrow{\mathbf{C}^{\text{T}}} \mathbf{h}_{\text{out}} \tag{31}$$

where $\mathbf{A}, \mathbf{B}, \mathbf{C} \in \mathbb{R}^{D \times d_s}$.

Theoretical Significance: The repeated application of matrix $\mathbf{A}$ enables modeling of higher-order state dependencies beyond first-order Markovian assumptions, analogous to multistep numerical integration in differential equations.

SSM Complexity (Three Branches):

$$\text{FLOPs}_{\text{SSM-total}} = 15LDd_s, \text{Params}_{\text{SSM-total}} = 12Dd_s + 6D \tag{32}$$

$$\text{SSM}_{\text{complexity}} = O\left(LDd_s\right) = O\left(L\right). \tag{33}$$

Architectural Distinction: The SSM processes sequences through matrix-vector products $\mathbb{R}^{L \times D} \times \mathbb{R}^{D \times d_s}$ with shared

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state parameters, avoiding the formation of $L \times L$ interaction matrices characteristic of attention mechanisms.

c) Adaptive multiscale fusion: The fusion mechanism [see (15)–(18)] employs learnable weighting with a two-layer bottleneck transformation ($3D \to 2D \to D$)

$$\text{FLOPs}_{\text{fusion}} = 8LD^2, \text{Params}_{\text{fusion}} = 8D^2 \tag{34}$$

d) Attention enhancement: The channel attention module [see (19) and (20)] utilizes a compression-expansion bottleneck ($D \to D/4 \to D$)

$$\text{FLOPs}_{\text{attention}} = \frac{LD^2}{2}, \text{Params}_{\text{attention}} = \frac{D^2}{2}. \tag{35}$$

Block Total Complexity:

$$\text{FLOPs}_{\text{block}} = 11.5LD^2 + 15LDd_s. \tag{36}$$

For typical configurations where $D \gg d_s$

$$\text{FLOPs}_{\text{block}} = O\left(LD^2\right) = O(L) \cdot O\left(D^2\right) \tag{37}$$

$$\text{Params}_{\text{block}} \approx 11.5D^2 + 12Dd_s. \tag{38}$$

Critical Observation: The complexity maintains linear dependence on sequence length L with a constant D^2 coefficient independent of spatial resolution.

The final projection layer transforms features to classification space

$$\text{FLOPs}_{\text{output}} = D^2L, \text{Params}_{\text{output}} = D^2. \tag{39}$$

2) Total Model Complexity: Cascading N blocks [see (24) and (25)]

$$\text{FLOPs}_{\text{total}} = O(CDL) + N \cdot O\left(LD^2\right) + O\left(D^2L\right) \tag{40}$$

$$\mathcal{T}_{\text{HierarchicalMamba}} = O\left(NLD^2\right) \tag{41}$$

$$\text{Params}_{\text{total}} \approx N\left(11.5D^2 + 12Dd_s\right). \tag{42}$$

The architecture exhibits linear scaling in sequence length L and quadratic scaling in model dimension D.

3) Comparison With Standard Transformer:

a) Self-attention mechanism: Standard transformer attention computes

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{softmax}\left(\frac{\mathbf{Q}\mathbf{K}^{\text{T}}}{\sqrt{d_k}}\right)\mathbf{V} \tag{43}$$

where $\mathbf{Q}, \mathbf{K}, \mathbf{V} \in \mathbb{R}^{L \times D}$.

b) Complexity analysis: Computing $\mathbf{K}^{\text{T}} \in \mathbb{R}^{L \times L}$: L^2D FLOPs

Attention-Weighted Aggregation: L^2DFLOPs

$$\mathcal{T}_{\text{Transformer}} = O\left(L^2D\right). \tag{44}$$

Computational Bottleneck: The construction of the $L \times L$ attention matrix necessitates all-pairs token interactions, resulting in quadratic scaling.

Complexity Ratio:

$$\frac{\mathcal{T}_{\text{HierarchicalMamba}}}{\mathcal{T}_{\text{Transformer}}} = \frac{NLD^2}{L^2D} = \frac{ND}{L}. \tag{45}$$

Scaling Analysis: The complexity ratio reveals a crossover point at $L = ND$, beyond which HierarchicalMamba achieves lower computational cost. The fundamental distinction lies in asymptotic behavior: HierarchicalMamba scales linearly ($\mathcal{T} \propto L$) while Transformer exhibits quadratic growth ($\mathcal{T} \propto L^2$). Consequently, doubling the sequence length incurs a $2\times$ increase for HierarchicalMamba versus $4\times$ for Transformers. As sequence length approaches values exceeding 1000 in high-resolution imagery, the ratio ND/L diminishes, establishing HierarchicalMamba's computational advantage for large-scale processing tasks.

Memory Requirements: HierarchicalMamba requires $O(BLD)$ memory compared to Transformer's $O(BLD + BL^2)$. When $L > D$, the $O(L^2)$ attention matrix storage dominates, making HierarchicalMamba substantially more memory-efficient for high-resolution inputs.

These derivations establish the linear scaling property of HierarchicalMamba for sequence processing tasks.

E. Algorithm Description

To provide a clear overview of the HierarchicalMamba framework, Algorithm 1 presents the complete computational procedure for HSI classification.

III. Experimental Dataset and Experimental Validation

A. Experimental Datasets and Evaluation Framework

This study conducts comprehensive evaluations on five representative HSI benchmark datasets that encompass diverse geographical environments, sensor characteristics, and classification challenges. The Indian Pines dataset was acquired over agricultural areas in northwestern Indiana, USA, featuring 145×145 spatial pixels and 200 spectral bands with 16 crop and vegetation categories. This dataset presents significant challenges due to high inter-class spectral similarity and severe class imbalance issues. The Pavia University dataset was captured by the reflective optics system imaging spectrometer sensor over the University of Pavia campus in Italy, comprising 610×340 spatial pixels, 103 spectral bands, and nine urban land cover classes, representing typical urban scene classification scenarios. The Salinas Valley dataset covers agricultural regions in Salinas Valley, CA, USA, with 512×217 spatial resolution and 204 spectral bands containing 16 crop categories, characterized by high spectral resolution and diverse agricultural classes. The Kennedy Space Center dataset encompasses wetland ecosystems at Kennedy Space Center, FL, USA, featuring 512×614 spatial pixels, 176 spectral bands, and 13 wetland categories, with challenges arising from complex ecological environments and subtle inter-class spectral differences. The Houston2018 dataset was acquired over the University of Houston campus and neighboring urban areas in Houston, TX, USA, comprising 4768×1202 spatial pixels with 48 spectral bands and 0.5 m spatial resolution. This dataset encompasses 20 urban land cover classes including healthy grass, stressed grass, artificial turf, evergreen trees, deciduous trees, bare earth, water, residential buildings, non-residential buildings, roads, sidewalks, crosswalks, major thoroughfares, highways, railways, paved parking lots, unpaved parking lots, cars, trains, and stadium seats, representing one of the most challenging

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Algorithm 1 HierarchicalMamba for HSI Classification

Input: HSI $\mathbf{X} \in \mathbb{R}^{B\times C\times H\times W}$, number of hierarchical blocks L, model dimension D, state dimension d_s
Output: Classification predictions $\mathbf{y} \in \mathbb{R}^{B\times N_{classes}}$

//Input Projection
$\mathbf{X}_{proj} \leftarrow$ Progressive-Conv($\mathbf{X}$)// (1)
//Spatial-Sequential Transformation
$\mathbf{X}_{seq} \leftarrow$ Reshape($\mathbf{X}, B, H \times W, D$)// (2), $L = H \times W$
$\mathbf{x}^{(0)} \leftarrow \mathbf{X}_{seq}$
//Hierarchical Processing
for l = 1to L **do**
  $\mathbf{x}_{norm} \leftarrow$ LayerNorm($\mathbf{x}^{(l|1)}$)
  //Multiscale Feature Extraction
  for i = 1to 3 **do**
  $\mathbf{f}_i \leftarrow$ Projection$_i$($\mathbf{x}_{norm}$)// (3)–(5)
  //Enhanced SSM with DST
  $\mathbf{g}_{state} \leftarrow \sigma$(Linear$_{d_{state}}$($\mathbf{f}_i$))// (6)
  $\mathbf{h} \leftarrow \mathbf{f}_i \odot \exp(\boldsymbol{\delta})$// (7)
  $\mathbf{h}_{state} \leftarrow \mathbf{hA}$/ First transition, (8)
  $\mathbf{h}_{\text{gated}} \leftarrow \mathbf{h}_{state} \odot \mathbf{g}_{state}$// (9)
  $\mathbf{h}_{proj} \leftarrow \mathbf{h}_{\text{gated}}\mathbf{B}^T$// (10)
  $\mathbf{h}_{state2} \leftarrow \mathbf{h}_{proj}\mathbf{A}$//Second transition, (11)
  $\mathbf{h}_{\text{out}} \leftarrow \mathbf{h}_{state2}\mathbf{C}^T$// (12)
  $\mathbf{h}_{\text{direct}} \leftarrow \mathbf{f}_i \odot \mathbf{D}$// (13)
  $\mathbf{f}_{\text{weighted},i} \leftarrow (\mathbf{h}_{\text{out}} + \mathbf{h}_{\text{direct}})\cdot\omega_i$//SSM output, (14)–(16)
end for
//Adaptive Multiscale Fusion
$\mathbf{f}_{\text{combined}} \leftarrow$ Concat($[\mathbf{f}_{\text{weighted},1}, \mathbf{f}_{\text{weighted},2}, \mathbf{f}_{\text{weighted},3}]$)// (17)
$\mathbf{f}_{\text{fused}} \leftarrow$ LN(Linear$_{d_{model}}$(GELU(LN(Linear$_{2d_{model}}$(($\mathbf{f}_{\text{combined}}$))))) // (18)
//Attention Enhancement
$\mathbf{A}_{\text{att}} \leftarrow \sigma$(Linear$_{d_{model}}$(GELU(Linear$_{d_{model}/4}$($\mathbf{f}_{\text{fused}}$))))// (19)
$\mathbf{f}_{enhanced} \leftarrow \mathbf{f}_{\text{fused}} \odot \mathbf{A}_{\text{att}}$// (20)
//Residual Connection
$\mathbf{x}^{(l)} \leftarrow \mathbf{x}^{(l-1)} + \mathbf{f}_{enhanced}$// (21)and (22)
end for
//Output Reconstruction and Classification
$\mathbf{X}_{final} \leftarrow$ Permute(Reshape($\mathbf{x}^{(L)}$))// (26)
$\mathbf{y} \leftarrow$ Classifier(GlobalAvgPool(Conv($\mathbf{X}_{final}$)))// (27)
return $\mathbf{y}$

HSI classification benchmarks due to high spatial resolution with fine-grained structural details, spectrally similar categories among pavement and building materials, severe class imbalance, and complex mixed pixels at land cover boundaries.

B. Comparative Methods and Baseline Configuration

Traditional convolutional neural network approaches include cnn_deep employing multi-layer 3-D convolutional structures [67], cnn_inception based on Inception module design [68], and cnn_efficientnetv2 featuring efficient lightweight architecture [69], representing typical applications of convolutional neural networks in HSI classification. Attention mechanism enhanced methods encompass ATN_cbam [70] combining channel and spatial attention, ATN_nonlocal [71] modeling long-range dependencies, ATN_spectral [72] specifically designed for spectral dimension processing, and ATN_dynamic adaptively adjusting attention weights, demonstrating diversified design approaches of attention mechanisms in hyperspectral feature learning. Transformer-series methods include pyramid vision Transformer with hierarchical design (TRANSF_pvt_tiny) [73], unified architecture (TRANSF_uniformer_tiny) [74], dual-branch structure (TRANSF_twins_tiny) [78], and lightweight mobile vision Transformer (TRANSF_mobilevit_tiny) [75], reflecting the application potential of sequence modeling techniques in HSI processing. Through comparison with these representative baseline methods, the relative advantages of HierarchicalMamba across different technical paradigms can be objectively evaluated.

C. Experimental Platform and Training Configuration

Experiments were conducted on a high-performance computing platform equipped with NVIDIA RTX 4090 GPU, utilizing PyTorch 2.0+ deep learning framework integrated with scientific computing libraries including NumPy, SciPy, and the Scikit-learn machine learning toolkit to construct a comprehensive experimental environment. Training configurations adopt unified parameter settings to ensure fair comparison, including maximum training epochs of 500, batch size of 16, initial learning rate of 0.001, and weight decay of 0.0005. The learning rate scheduling strategy employed cosine annealing with warmup, setting warmup epochs to 5, minimum learning rate to 1×10^{-7}, and warmup starting learning rate to 1×10^{-6}, enabling stable gradient updates during early training phases and smooth learning rate decay in later stages. Early stopping mechanism configured patience of 50 epochs with minimum improvement threshold of 1×10^{-6}, monitoring validation loss changes to prevent overfitting. Data preprocessing utilized 7×7 pixel patch sizes for spatial neighborhood information extraction, applied reflective padding for boundary handling, and performed Z-score normalization across all data to eliminate scale differences among spectral bands. Dataset-specific training samples are set to 8% for Indian Pines and Houston2018, and 4% for Pavia University, Salinas Valley, and Kennedy Space Center.

To facilitate experimental replication, we provide key implementation specifications. The multiscale fusion weight parameter is initialized as torch.ones (3) with softmax normalization (exponential normalization ensuring non-negativity and unit sum) during forward propagation. The DST mechanism employs parameter sharing—the same matrix A ($d_{\text{model}} \times d_{\text{state}}$) is applied twice within each computational cycle, ensuring consistent dynamics while maintaining parameter efficiency.

State space components use the following initialization: state dimension $d_{\text{state}} = 16$; matrix A with normal distribution (mean = −0.5, std = 0.1); matrices B and C with normal distribution (mean = 0.0, std = 0.1); direct term D with normal distribution (mean = 1.0, std = 0.1); temporal step δ with normal distribution (mean = 0.0, std = 0.1). Linear layers employ Xavier uniform initialization for weights and zero initialization for biases. Convolutional layers use

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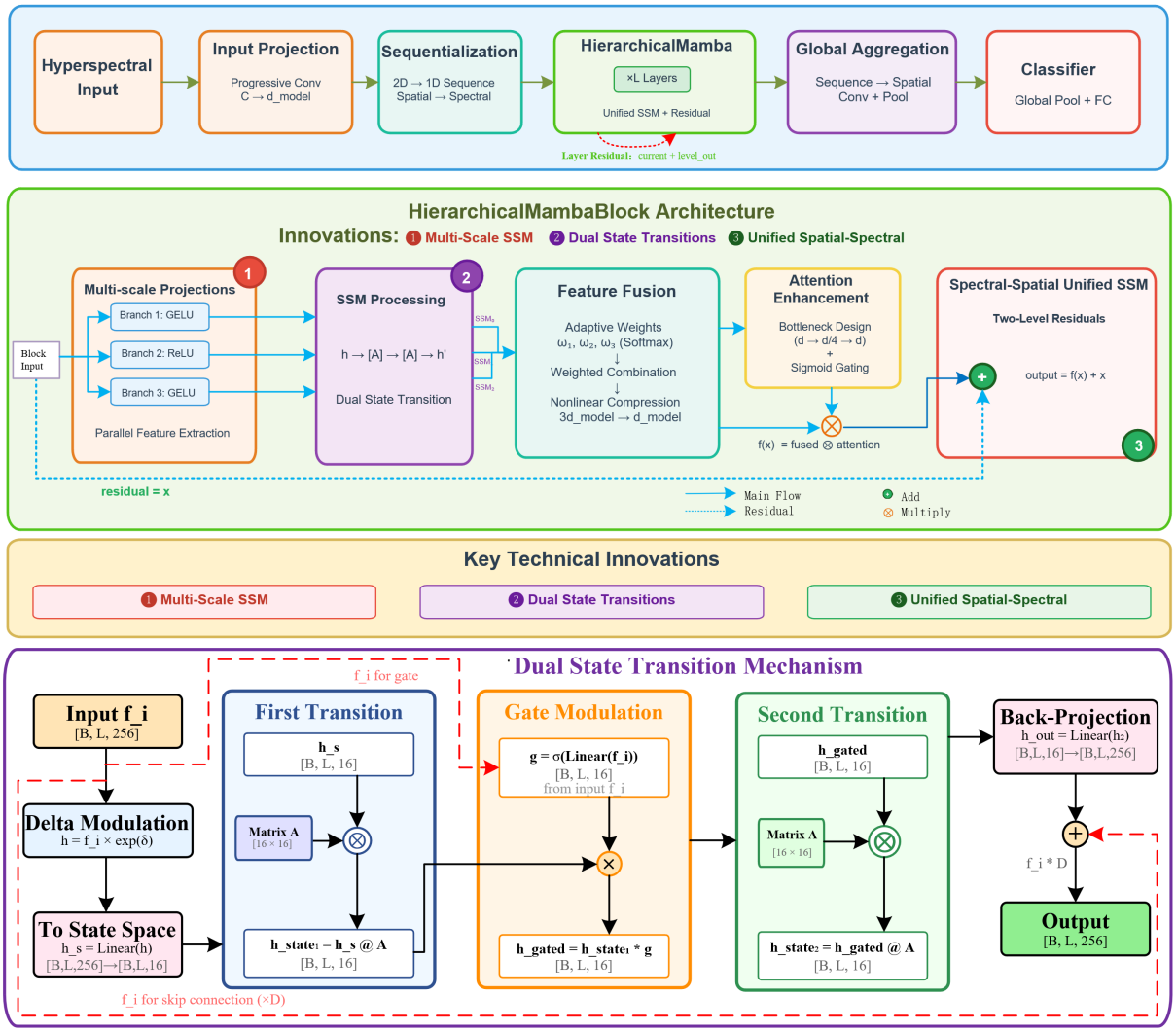


Fig. 1. HierarchicalMamba HSI classification network architecture.

Kaiming normal initialization (fan-out mode, ReLU nonlinearity). Layer normalization employs eps = $1e^{-5}$ with learnable affine parameters (weights initialized to ones, biases to zeros).

Architectural specifications include $d_{\text{model}} = 256$ and num_levels = 3. The three parallel branches maintain identical projection dimensions with distinct activations (GELU for branches 1 and 3, ReLU for branch 2). Dropout rate of 0.5 is applied after each HierarchicalMambaBlock during training. The Adam optimizer uses default hyperparameters (betas = (0.9, 0.999), eps = $1e^{-8}$).

Gradient clipping employs maximum norm 1.0. Random seed 42 ensures reproducibility with float32 precision throughout.

Dataset-specific training configurations were differentiated based on individual dataset characteristics: Indian Pines and Houston2018 employed 8% training samples to address class imbalance issues, while Pavia University, Salinas Valley, and Kennedy Space Center utilized 4% training samples. This configuration considers both dataset scale differences and ensures model generalization capability evaluation under small-sample learning scenarios. The optimization strategy employed Adam optimizer combined with cross-entropy loss function, implementing gradient clipping mechanisms to prevent gradient explosion. Model checkpoint management adopted a strategy of saving every 5 epochs, automatically preserving the best model based on validation performance, retaining a maximum of three checkpoint files to balance storage efficiency and model recovery capability. The evaluation framework encompasses standard metrics including overall accuracy (OA), average accuracy (AA), Kappa coefficient, as well as class-specific precision, recall, and confusion matrix analysis, providing comprehensive quantitative assessment of algorithm performance.

D. Experiments and Analysis

1) Experiment of Indian Pines: The Indian Pines dataset, recognized as a canonical benchmark in HSI classification, is characterized by diverse agricultural crop categories and complex spectral features. As shown in Table III, HierarchicalMamba achieved significant performance improvements on this dataset, obtaining 96.11% OA, 91.56% AA, and

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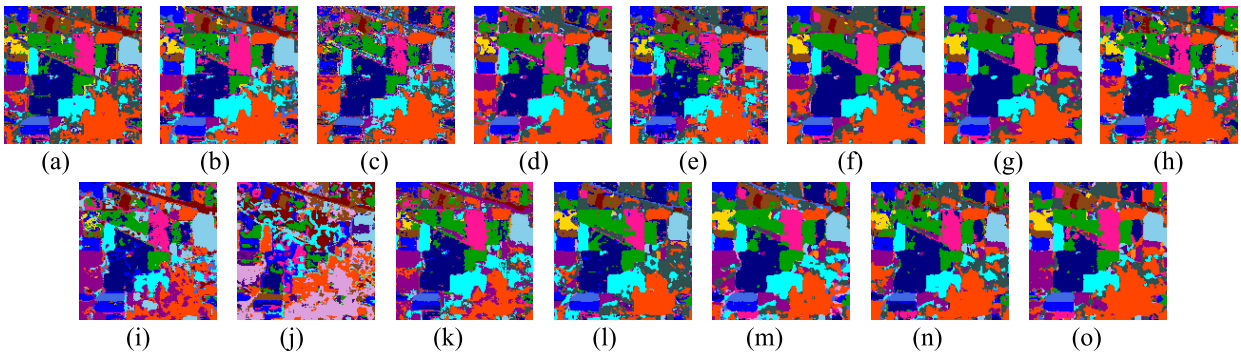


Fig. 2. Classification map on the Indian Pines dataset: (a) cnn_deep, (b) cnn_inception, (c) cnn_efficientnetv2, (d) ATN_cbam, (e) ATN_nonlocal, (f) ATN_spectral, (g) ATN_dynamic, (h) TRANSF_pvt_tiny, (i) TRANSF_uniformer_tiny, (j) TRANSF_twins_tiny, (k) TRANSF_mobilevit_tiny, (l) MSDANet, (m) DualPathmamba, (n) HyperspectralMamba, and (o) HierarchicalMamba.

TABLE III

COMPARISON OF CLASSIFICATION ACCURACIES (%) PROVIDED BY DIFFERENT METHODS (INDIAN PINES DATASET)

Ground	cnn_deep	cnn_inception	cnn_efficientnetv2	ATN_cbam	ATN_nonlocal	ATN_spectral	ATN_dynamic	TRANSF_pvt_tiny	TRANSF_uniformer_tiny	TRANSF_twins_tiny	TRANSF_mobilevit_tiny	MSDANet	DualPathmamba	HyperspectralMamba	HierarchicalMamba
Alfalfa	50.00	30.95	0.00	21.43	33.33	42.86	69.05	0.00	9.52	35.71	33.33	20.00	17.50	12.50	45.24
Corn-no-till	76.56	88.74	62.94	87.82	84.32	93.00	92.16	85.46	86.23	72.15	76.41	96.25	98.25	88.67	97.49
Corn-min-till	76.57	90.05	39.53	73.82	73.17	85.34	85.73	79.84	79.06	80.24	50.92	93.55	89.11	65.19	90.31
Corn	93.58	80.73	47.25	98.62	71.56	99.54	98.17	91.28	91.74	83.49	92.66	91.04	90.55	76.62	93.12
Grass-pasture	90.77	92.34	81.08	94.14	88.96	92.57	94.82	94.82	95.05	89.19	84.46	98.03	98.77	95.82	96.17
Grass-trees	97.47	99.26	98.07	96.28	93.90	97.32	98.21	97.92	99.70	98.51	94.20	99.02	98.53	96.42	99.40
Grass-pasture-mowed	84.62	96.15	0.00	57.69	96.15	96.15	92.31	92.31	76.92	76.92	65.38	90.91	95.45	72.73	61.54
Hay-windrowed	100.00	100.00	99.32	99.32	93.64	98.41	96.36	33.41	100.00	96.14	100.00	100.00	100.00	98.51	100.00
Oats	38.89	50.00	0.00	44.44	44.44	38.89	44.44	100.00	72.22	88.89	44.44	100.00	71.43	35.71	100.00
Soybeans-no-till	82.55	71.92	37.02	91.28	90.38	90.27	90.27	79.19	46.64	92.17	78.52	72.37	87.29	82.76	94.74
Soybeans-min-till	90.66	95.35	64.63	88.53	89.29	97.03	97.21	92.61	95.44	78.22	95.93	87.40	94.09	94.67	97.88
Soybeans-clean-till	69.96	82.23	39.56	74.73	92.31	96.15	96.52	69.96	76.01	17.58	40.48	85.97	85.57	37.88	96.15
Wheat	100.00	100.00	93.65	100.00	100.00	100.00	100.00	95.24	99.47	100.00	98.41	100.00	100.00	99.42	100.00
Woods	90.98	95.70	40.89	96.48	98.28	99.57	98.20	94.76	76.89	90.89	99.66	94.45	93.70	99.06	93.81
Bldg-grass-tree	93.52	78.31	52.68	83.66	92.96	93.24	91.83	84.51	85.63	91.55	63.94	88.65	88.96	79.45	99.15
Stone-steel-towers	100.00	100.00	75.29	100.00	95.29	100.00	98.82	77.65	95.29	100.00	100.00	98.73	98.73	87.34	100.00
OA	86.70	90.20	59.37	88.92	88.88	94.67	94.57	85.16	84.43	80.80	83.03	90.61	93.39	86.43	**96.11**
AA	83.51	84.48	51.99	81.77	83.62	88.77	90.26	79.31	80.36	80.73	76.17	88.52	88.00	76.42	**91.56**
Kappa	84.82	88.80	53.83	87.37	87.32	93.92	93.80	83.05	82.20	78.17	80.33	89.32	92.46	84.34	**95.57**

TABLE IV

COMPARISON OF CLASSIFICATION ACCURACY (%) PROVIDED BY DIFFERENT METHODS (PAVIA UNIVERSITY DATASET)

Class Name	cnn_deep	cnn_inception	cnn_efficientnetv2	ATN_cbam	ATN_nonlocal	ATN_spectral	ATN_dynamic	TRANSF_pvt_tiny	TRANSF_uniformer_tiny	TRANSF_twins_tiny	TRANSF_mobilevit_tiny	MSDANet	DualPathmamba	HyperspectralMamba	HierarchicalMamba
Asphalt	99.01	98.18	94.86	97.74	93.73	98.02	98.90	97.03	97.85	96.15	93.76	95.56	97.72	95.25	98.95
Meadows	99.06	99.96	98.47	98.84	98.78	98.26	99.41	97.18	99.38	99.49	99.78	99.85	99.65	84.45	99.97
Gravel	87.79	90.47	83.62	37.92	62.88	63.97	56.97	85.26	47.34	78.31	80.94	86.24	99.12	61.15	97.67
Trees	96.09	96.67	94.49	89.66	97.04	98.64	96.16	92.96	95.65	98.88	90.99	96.10	97.80	97.87	98.33
Metal Sheets	100.00	99.85	99.92	95.89	99.77	99.46	99.54	100.00	98.37	100.00	100.00	100.00	100.00	98.71	99.92
Bare Soil	96.25	88.11	92.23	93.64	93.25	99.73	93.64	99.01	70.46	96.13	94.70	92.87	98.88	99.03	92.90
Bitumen	90.99	93.27	80.89	75.57	86.22	92.25	85.90	97.65	80.42	97.10	95.46	97.47	97.39	72.30	99.14
Bricks	94.99	94.12	90.44	89.34	96.75	95.28	97.85	91.20	98.84	95.84	96.72	97.79	72.20	98.32	91.97
Shadows	99.12	99.56	99.12	97.47	98.68	98.68	99.67	97.69	93.29	98.90	100.00	100.00	100.00	99.89	99.78
OA	97.38	96.86	94.99	92.75	94.92	96.34	95.79	96.08	92.12	97.11	96.31	97.18	96.69	89.26	**98.03**
AA	95.92	95.57	92.67	86.23	91.90	93.81	92.01	95.33	86.85	95.64	94.71	96.21	95.86	89.66	**97.62**
Kappa	96.53	95.82	93.34	90.32	93.27	95.16	94.40	94.83	89.39	96.17	95.09	96.25	95.61	86.14	**97.38**

95.57% Kappa coefficient, representing the highest performance among all comparative methods. Specifically, the algorithm demonstrated superior performance across the majority of the 16 land cover categories, particularly achieving 97.49% classification accuracy for the Corn-no-till category, representing a 4.49 percentage point improvement over the second-best method. In particular, HierarchicalMamba achieved perfect 100% classification accuracy on the challenging small-sample Oats category, significantly outperforming other methods that generally achieved less than 50% accuracy on this category. This superior performance reveals the algorithm's effectiveness in addressing class imbalance

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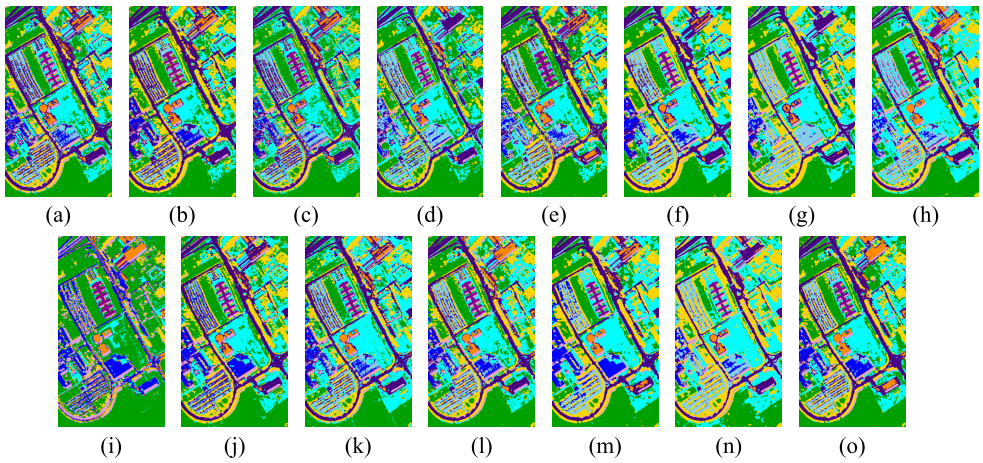


Fig. 3. Classification map on the Pavia University dataset: (a) cnn_deep, (b) cnn_inception, (c) cnn_efficientnetv2, (d) ATN_cbam, (e) ATN_nonlocal, (f) ATN_spectral, (g) ATN_dynamic, (h) TRANSF_pvt_tiny, (i) TRANSF_uniformer_tiny, (j) TRANSF_twins_tiny, (k) TRANSF_mobilevit_tiny, (l) MSDANet, (m) DualPathmamba, (n) HyperspectralMamba, and (o) HierarchicalMamba.

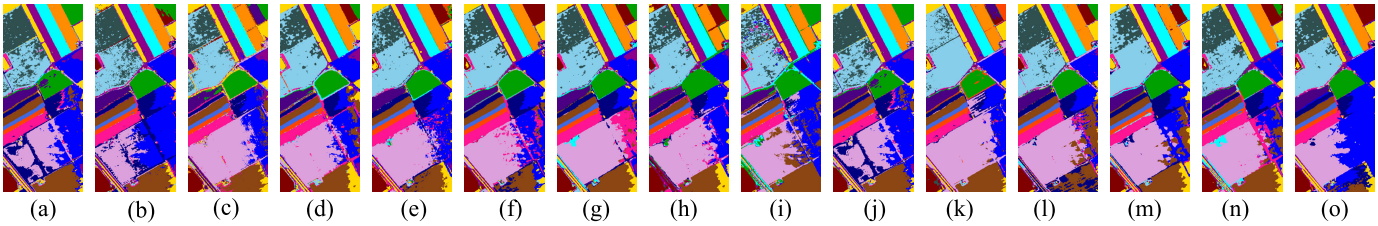


Fig. 4. Classification map on the Salinas Valley dataset: (a) cnn_deep, (b) cnn_inception, (c) cnn_efficientnetv2, (d) ATN_cbam, (e) ATN_nonlocal, (f) ATN_spectral, (g) ATN_dynamic, (h) TRANSF_pvt_tiny, (i) TRANSF_uniformer_tiny, (j) TRANSF_twins_tiny, (k) TRANSF_mobilevit_tiny, (l) MSDANet, (m) DualPathmamba, (n) HyperspectralMamba, and (o) HierarchicalMamba.

issues. The classification map presented in Fig. 2 illustrates that HierarchicalMamba produced results with excellent spatial consistency and well-defined boundaries, effectively suppressing the salt-and-pepper noise phenomena commonly observed in traditional methods, while maintaining superior spatial structure preservation compared to baseline approaches.

2) Experiment of Pavia University: The Pavia University dataset is renowned for its urban scene complexity and high spatial resolution characteristics, imposing stringent requirements on algorithms' spatial information utilization capabilities. Experimental results present that HierarchicalMamba maintained competitive performance on this dataset, achieving 98.03% OA, 97.62% AA, and 97.38% Kappa coefficient, ranking first among all comparative algorithms (see Table IV). From a category-level analysis perspective, the algorithm achieved above 95% classification accuracy on 8 out of 9 land cover categories, with particularly outstanding performance on the technically challenging Gravel category, obtaining 97.67% accuracy. This result significantly surpasses other methods' performance, with most achieving below 90% accuracy on this category. This superior performance is primarily attributed to HierarchicalMamba's multiscale feature decomposition mechanism, which effectively captures subtle spectral differences among different among different land cover categories.

Fig. 3 further validates the algorithm's advantages, demonstrating HierarchicalMamba's outstanding performance in maintaining spatial structure integrity, accurately identifying boundaries. Between different land cover categories while avoiding boundary blurring and misclassification phenomena.

3) Experiment of Salinas Valley: The Salinas Valley dataset is distinguished by its rich crop diversity and high spectral resolution, serving as an important benchmark for evaluating algorithms' spectral discrimination capabilities. In experiments on this dataset, HierarchicalMamba manifested consistent performance, achieving 97.06% OA, 98.80% AA, and 96.73% Kappa coefficient, again obtaining optimal performance among all comparative methods. Detailed category-wise analysis results in Table V reveal that the algorithm achieved perfect 100% classification accuracy on 10 out of 16 crop categories, including outstanding performance on spectrally similar categories such as Broccoli-green-weeds-1 and Broccoli-green-weeds-2. Particularly noteworthy is HierarchicalMamba's achievement of 95.56% accuracy on the challenging Vineyard-untrained category, representing significant improvement over other methods (most achieving below 85% accuracy). This performance conclusively demonstrates the effectiveness of the DST mechanism in processing complex spectral patterns. The classification results displayed in Fig. 4 exhibit excellent spatial continuity and category discrimination, with clearly delineated crop cultivation area boundaries, further validating the algorithm's practical value.

4) Experiment on Kennedy Space Center: The Kennedy Space Center dataset represents the complex spectral envi-

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TABLE V

COMPARISON OF CLASSIFICATION ACCURACIES (%) PROVIDED BY DIFFERENT METHODS (SALINAS VALLEY DATASET)

Ground	cnn_deep	cnn_inception	cnn_efficientnetv2	ATN_cbam	ATN_nonlocal	ATN_spectral	ATN_dynamic	TRANSF_pvt_tiny	TRANSF_uniformer_tiny	TRANSF_twins_tiny	TRANSF_mobilevit_tiny	MSDANet	DualPathmamba	HyperspectralMamba	HierarchicalMamba
Broccoli-green-weeds-1	95.23	100.00	12.55	99.69	99.22	99.95	100.00	99.95	100.00	99.95	100.00	99.84	100.00	99.95	100.00
Broccoli green-weeds-2	100.00	99.69	99.75	99.97	100.00	100.00	100.00	99.78	52.42	100.00	100.00	100.00	100.00	99.68	100.00
Fallow	99.95	99.89	98.68	98.89	97.05	99.95	100.00	100.00	99.79	99.95	100.00	99.61	100.00	97.47	100.00
Fallow-rough-plough	99.85	100.00	99.78	98.06	97.91	98.88	98.58	98.65	99.70	97.76	98.21	100.00	99.69	99.38	99.85
Fallow-smooth	96.73	89.89	97.55	97.86	98.37	98.44	97.82	97.55	89.73	81.45	99.03	98.94	99.11	99.84	98.72
Stubble	100.00	100.00	99.97	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00	100.00
Celery	99.45	89.49	98.60	98.86	99.71	99.83	99.83	96.07	95.98	99.94	99.97	99.97	97.54	99.79	100.00
Grapes-untrained	86.67	90.26	71.14	85.25	95.55	86.89	91.40	93.41	98.70	92.74	94.56	88.26	98.76	86.70	89.52
Soil-vineyard-develop	99.68	99.51	99.95	99.78	100.00	100.00	100.00	99.95	93.89	99.92	99.95	99.95	100.00	99.98	100.00
Corn-senesced-green weeds	98.73	98.79	94.47	93.33	93.49	99.46	98.38	96.41	98.82	99.08	98.51	99.27	95.82	99.47	99.40
Lettuce-romaine-4wk	94.63	90.93	94.34	95.80	99.02	98.34	98.34	98.93	88.78	96.78	97.66	100.00	99.29	97.97	99.41
Lettuce-romaine-5wk	99.95	100.00	100.00	99.95	99.19	99.73	100.00	85.73	98.70	100.00	100.00	100.00	100.00	98.20	100.00
Lettuce-romaine-6wk	99.54	98.52	99.77	100.00	97.84	99.43	97.38	97.27	95.11	97.84	100.00	100.00	100.00	99.88	99.89
Lettuce-romaine-7wk	95.91	98.73	95.72	96.20	96.59	96.98	95.62	98.34	89.78	98.44	97.37	98.78	97.67	93.41	99.32
Vineyard-untrained	85.29	76.84	92.27	83.50	47.27	92.33	87.07	68.85	46.80	74.03	78.79	94.14	60.00	90.76	95.56
Vineyard-vertical-trellis	98.44	96.60	95.97	98.73	99.08	99.08	98.44	98.79	98.56	99.02	99.08	98.32	98.74	97.47	99.14
OA	94.51	93.17	88.74	93.80	91.13	95.95	96.00	93.14	87.15	93.79	95.70	96.56	93.80	95.48	**97.06**
AA	96.88	95.57	90.66	96.62	95.02	98.08	97.68	95.61	90.42	96.06	97.69	98.57	96.66	97.50	**98.80**
Kappa	93.89	92.39	87.50	93.10	90.07	95.49	95.55	92.35	85.64	93.08	95.20	96.18	93.07	94.97	**96.73**

TABLE VI

COMPARISON OF CLASSIFICATION ACCURACY (IN PERCENT) PROVIDED BY DIFFERENT METHODS (KENNEDY SPACE CENTER DATASET)

Ground	cnn_deep	cnn_inception	cnn_efficientnetv2	ATN_cbam	ATN_nonlocal	ATN_spectral	ATN_dynamic	TRANSF_pvt_tiny	TRANSF_uniformer_tiny	TRANSF_twins_tiny	TRANSF_mobilevit_tiny	MSDANet	DualPathmamba	HyperspectralMamba	HierarchicalMamba
Scrub	98.50	99.59	96.85	99.45	95.35	100.00	98.08	98.91	99.45	98.77	99.45	78.89	98.29	99.00	100.00
Swamp willow	69.96	93.56	75.11	87.12	80.69	93.13	93.99	91.42	93.56	82.40	83.26	89.33	87.11	94.67	97.00
Cabbage palm hammock	0.41	89.43	77.24	85.37	65.45	97.97	97.97	48.37	99.19	95.53	100.00	55.51	100.00	97.03	99.19
Cabbage palm/oak	0.41	60.33	47.93	50.83	50.41	65.70	57.02	74.38	9.09	51.65	12.40	42.67	43.97	40.09	80.58
Slash pine	1.30	68.83	33.12	28.57	91.56	85.71	81.82	0.00	28.57	20.78	3.25	85.91	67.11	6.71	81.17
Oak/broadleaf hammock	75.45	44.09	50.00	84.55	91.36	75.45	82.73	42.73	62.27	40.91	5.45	9.48	62.09	72.99	57.73
Hardwood swamp	100.00	76.24	20.79	97.03	87.13	100.00	99.01	51.49	98.02	100.00	86.14	50.52	100.00	100.00	79.21
Graminoid marsh	83.33	66.67	48.07	93.96	99.52	95.17	91.30	80.92	89.37	80.43	86.71	90.93	99.24	87.91	98.31
Spartina marsh	95.99	98.40	96.79	99.00	87.37	100.00	100.00	79.56	93.79	97.19	97.80	100.00	98.75	100.00	100.00
Cattail marsh	98.97	97.42	73.45	85.31	96.39	93.56	94.85	99.23	93.30	94.59	93.81	86.83	100.00	81.45	97.42
Salt marsh	97.01	96.77	90.55	98.26	99.50	99.75	100.00	97.26	100.00	99.75	94.28	96.90	100.00	100.00	98.01
Muld flats	69.98	97.93	79.09	86.13	94.82	94.82	97.52	89.65	89.23	93.17	89.44	48.60	100.00	87.04	98.14
Water	100.00	100.00	100.00	100.00	100.00	100.00	100.00	84.61	98.76	100.00	100.00	100.00	100.00	100.00	100.00
OA	79.55	89.73	79.43	90.07	91.31	94.98	94.56	81.45	88.01	88.41	84.21	79.08	93.57	88.80	**95.34**
AA	68.56	83.79	68.38	84.27	87.66	92.41	91.87	72.19	81.12	81.17	73.23	71.97	88.97	82.07	**91.29**
Kappa	77.26	88.53	77.08	88.93	90.32	94.41	93.95	79.36	86.63	87.08	82.36	76.68	92.85	87.51	**94.80**

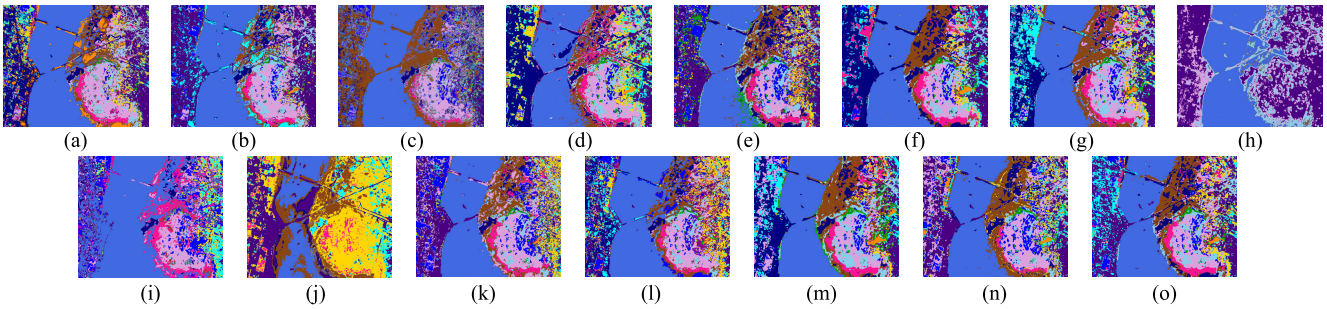


Fig. 5. Classification map on the Kennedy Space Center: (a) cnn_deep, (b) cnn_inception, (c) cnn_efficientnetv2, (d) ATN_cbam, (e) ATN_nonlocal, (f) ATN_spectral, (g) ATN_dynamic, (h) TRANSF_pvt_tiny, (i) TRANSF_uniformer_tiny, (j) TRANSF_twins_tiny, (k) TRANSF_mobilevit_tiny, (l) MSDANet, (m) DualPathmamba, (n) HyperspectralMamba, and (o) HierarchicalMamba.

ronment of wetland ecosystems, where relatively small spectral differences between land cover categories pose severe challenges to algorithms' fine discrimination capabilities. Experimental results show that HierarchicalMamba maintained superior performance on this dataset, achieving 95.34% OA, 91.29% AA, and 94.80% Kappa coefficient, continuing to rank first among all comparative methods. Detailed analysis of Table VI results reveals that the algorithm performed excellently across the majority of the 13 wetland categories, particularly achieving perfect 100% classification

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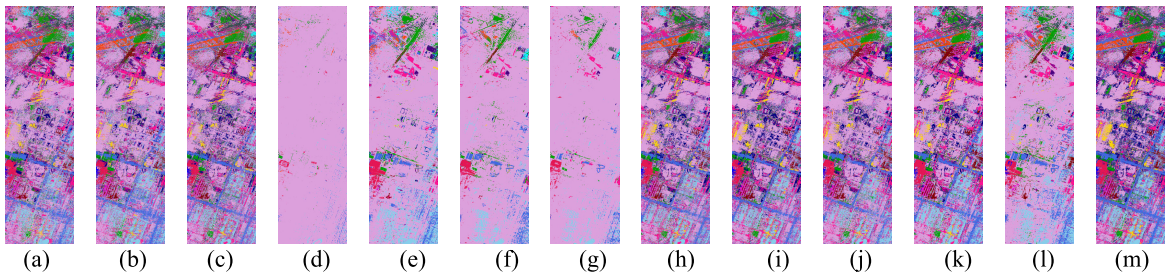


Fig. 6. Classification map on the Houston2018 dataset: (a) cnn_deep, (b) cnn_inception, (c) cnn_efficientnetv2, (d) ATN_cbam, (e) ATN_nonlocal, (f) ATN_spectral, (g) ATN_dynamic (h), TRANSF_pvt_tiny, (i) TRANSF_mobilevit_tiny, (j) MSDANet, (k) DualPathmamba, (l) HyperspectralMamba, and (m) HierarchicalMamba.

accuracy on Scrub, Spartina marsh, and Water categories. Particularly significant is HierarchicalMamba's achievement of 99.19% accuracy on the Cabbage palm hammock category, where most comparative methods achieved below 50% accuracy. This substantial performance difference validates the advantages of the unified spectral-spatial modeling framework.

The classification results shown in Fig. 5 exhibit clear spatial pattern distribution with minimal misclassification phenomena, accurately identifying and distinguishing the complex structures of wetland ecosystems.

5) Experiment of Houston2018: The Houston2018 dataset represents a significantly more challenging benchmark for HSI classification, characterized by high spatial resolution (0.5 m), 20 land cover categories, and substantial class imbalance in complex urban environments. As shown in Table VII, the algorithm significantly outperforms traditional CNN-based approaches (cnn_deep: 66.87% OA, cnn_inception: 69.51% OA, cnn_efficientnetv2: 65.72% OA) and achieves competitive performance compared to state-of-the-art Mamba variants (DualPathmamba: 80.17% OA, HyperspectralMamba: 68.66% Kappa). Category-level analysis reveals exceptional performance on Non-residential buildings (95.73%), Stadium seats (92.82%), Railways (87.25%), and Highways (83.64%), demonstrating superior capability in distinguishing diverse urban land cover types. The full-plot classification map in Fig. 6(m), including background regions, validates the model's robust generalization without overfitting artifacts, exhibiting excellent spatial continuity in linear features, coherent building classification patterns, and accurate boundary delineation between land cover transitions. While performance on challenging categories such as Crosswalks (28.00%) and Sidewalks (44.53%) remains limited due to extremely small spatial extents and severe spectral similarity with surrounding materials, HierarchicalMamba demonstrates consistent advantages over baseline methods, particularly excelling in complex building-related and transportation infrastructure categories that benefit from the DST mechanism and multiscale SSM decomposition.

6) Comparative Analysis of Classification Capabilities: Through comprehensive horizontal comparison of all baseline methods across the five datasets, distinct advantages and limitations of different technical approaches can be clearly identified. Traditional CNN methods (cnn_deep, cnn_inception, cnn_efficientnetv2) exhibited pronounced instability across all datasets, with cnn_inception demonstrating relatively stable performance but still showing apparent limitations when processing complex spectral features, particularly in distinguishing small-sample categories and spectrally similar categories.

a) Traditional convolutional neural network performance analysis: Traditional CNN methods exhibited significant performance variations and instability across the five datasets. Specifically, the cnn_deep method achieved 86.70% OA on Indian Pines, 97.38% on Pavia University, 94.51% on Salinas Valley, 79.55% on Kennedy Space Center, and 66.87% on Houston2018, with a high standard deviation of 11.28%, reflecting poor adaptability to different data characteristics.

The cnn_inception method reflected relatively stable performance, with OAs of 90.20%, 96.86%, 93.17%, 89.73%, and 69.51% across the five datasets respectively, achieving a standard deviation of 10.79% and representing the best performance among traditional CNN methods.

The cnn_efficientnetv2 displayed the most unstable performance, with OA ranging from 59.37% on Indian Pines to 94.99% on Pavia University, yielding a standard deviation of 14.35%.

b) Attention mechanism enhanced methods performance evaluation: Attention-enhanced CNN methods demonstrated clear performance improvements compared to traditional CNNs. The ATN_cbam method achieved OAs of 88.92%, 92.75%, 93.80%, 90.07%, and 44.73% across the five datasets, with AA of 82.05% and standard deviation of 19.91%.

ATN_spectral exhibited the most outstanding performance among attention mechanisms, achieving OAs of 94.67%, 96.34%, 95.95%, 94.98%, and 47.95% across the five datasets respectively, with AA of 86.00% and standard deviation of 21.68% (primarily due to the challenging Houston2018 dataset).

ATN_dynamic similarly performed excellently, achieving an average OA of 86.04%, particularly obtaining 96.00% high accuracy on Salinas Valley.

c) Transformer series methods performance analysis: Transformer series methods manifested sequential modeling potential but exhibited obvious performance fluctuations overall. TRANSF_pvt_tiny showed significant variations across the five datasets, with OA ranging from 75.82% on Hous-

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TABLE VII
COMPARISON OF CLASSIFICATION ACCURACIES (%) PROVIDED BY DIFFERENT METHODS (HOUSTON2018 DATASET)

Ground	cnn_deep	cnn_inception	cnn_efficientnetv2	ATN_cbam	ATN_nonlocal	ATN_spectral	ATN_dynamic	TRANSF_pvt_tiny	TRANSF_mobilevit_tiny	MSDANet	DualPathmamba	HyperspectralMamba	HierarchicalMamba
Healthy grass	48.95	46.59	46.99	0.74	14.12	5.47	9.31	59.74	66.47	55.79	71.14	36.69	75.92
Stressed grass	56.09	65.75	58.46	0.16	35.24	14.00	16.11	68.96	74.09	66.50	75.95	56.29	75.54
Artificial turf	59.35	63.44	61.00	3.74	1.75	12.47	0.00	78.30	75.61	64.19	85.84	72.62	79.56
Evergreen trees	47.13	54.43	48.91	0.00	5.10	1.01	0.48	65.93	71.40	60.10	79.51	37.36	79.85
Deciduous trees	29.63	39.99	33.91	0.00	5.56	0.21	0.17	49.14	59.62	50.95	64.56	33.04	62.54
Bare earth	57.05	70.14	54.12	0.00	26.02	1.32	0.76	70.02	77.38	73.29	79.83	80.84	86.15
Water	34.70	41.74	41.74	0.00	0.00	0.00	0.00	49.42	68.76	69.65	66.07	63.12	63.30
Residential buildings	56.88	63.30	58.35	1.64	33.36	11.94	17.58	72.20	75.37	70.46	82.62	59.00	77.86
Non-residential buildings	89.13	88.30	85.35	99.76	91.93	98.16	97.77	90.02	89.84	88.38	90.97	91.11	95.73
Roads	39.75	46.33	44.11	0.35	5.52	4.05	3.08	61.35	63.51	48.61	59.77	20.95	70.83
Sidewalks	23.86	23.61	28.05	0.00	2.01	0.11	0.41	45.91	50.20	31.81	44.48	5.88	44.53
Crosswalks	8.20	19.12	15.00	0.00	0.00	0.00	0.00	44.47	47.12	23.63	33.13	6.83	28.00
Major thoroughfares	61.61	62.52	57.29	1.84	37.84	12.29	12.62	68.80	74.58	65.73	82.03	63.39	73.05
Highways	55.48	66.86	58.02	0.34	23.46	0.80	1.78	69.54	77.64	65.14	83.88	50.96	83.64
Railways	65.56	71.70	61.14	0.00	38.28	8.20	8.55	73.76	75.60	72.59	86.81	68.66	87.25
Paved parking lots	51.79	58.19	45.34	0.14	8.34	2.17	1.48	64.55	68.78	60.93	69.48	45.37	73.69
Unpaved parking lots	45.86	72.81	53.43	0.00	0.95	0.00	0.00	71.63	73.52	75.89	95.27	54.37	81.24
Cars	33.11	46.54	40.68	0.00	17.50	0.78	2.47	58.89	58.87	47.14	66.93	32.89	71.46
Trains	50.55	64.19	52.53	0.01	26.59	3.59	3.88	65.06	75.89	61.69	72.94	53.85	77.28
Stadium seats	81.45	83.74	79.09	10.68	78.03	52.06	45.30	86.89	90.89	88.75	94.50	85.23	92.82
OA	66.87	69.51	65.72	44.73	53.20	47.95	48.29	75.82	78.24	71.80	80.17	64.84	**82.35**
AA	49.81	57.46	51.17	5.97	22.58	11.43	11.09	65.73	70.76	62.06	74.28	50.92	**74.01**
Kappa	54.69	59.00	54.01	1.38	28.70	11.74	13.05	67.86	71.40	62.45	73.94	50.78	**76.46**

ton2018 to 96.08% on Pavia University, yielding a standard deviation of 8.32%.

TRANSF_uniformer_tiny reflected relative instability across four datasets (Indian Pines, Pavia University, Salinas Valley, and Kennedy Space Center), with OA ranging from 84.43% to 92.12%, but could not complete evaluation on Houston2018 due to GPU memory constraints.

TRANSF_twins_tiny exhibited moderate stability with an average OA of 90.03% across four datasets (Indian Pines, Pavia University, Salinas Valley, Kennedy Space Center), but could not complete evaluation on Houston2018 due to GPU memory constraints.

TRANSF_mobilevit_tiny sought balance between computational efficiency and performance, achieving an average OA of 87.51% across five datasets.

The comparative analysis across five diverse datasets reveals distinct performance hierarchies among methodological approaches. Traditional CNN methods and attention-enhanced mechanisms, while achieving strong performance on traditional benchmarks, exhibit substantial degradation on the challenging Houston2018 dataset (performance drops exceeding 25–48 percentage points). Transformer approaches show moderate dataset sensitivity with performance variations of 20–22 percentage points between easiest and most challenging scenarios. HierarchicalMamba demonstrates superior cross-dataset generalization, achieving 95.34%–98.03% accuracy on four traditional benchmarks while maintaining 82.35% on the challenging Houston2018 urban scene—representing only a 15.68 percentage point performance gap. Among all comparative methods, HierarchicalMamba achieves the smallest performance variation while maintaining consistently high accuracy across all datasets, validating the effectiveness of the unified spatial-sequential modeling framework for diverse hyperspectral imaging applications.

7) Ablation Study: To validate the contribution of each architectural component, we conducted comprehensive ablation experiments by systematically removing individual modules. Table VIII presents ablation results across three representative datasets: Indian Pines, Kennedy Space Center, and Houston2018.

DST (wo_dual): Removal caused consistent degradation with OA decreasing by 0.43%, 1.59%, and 3.07% across the three datasets respectively. The substantial 3.07% OA drop on Houston2018 shows this mechanism's importance for complex urban scenes, validating that dual application of the state transition matrix enhances modeling of higher-order state dependencies and nonlinear dynamics.

Direct Term D (wo_direct): Ablating the feedthrough term resulted in 0.40%, 1.17%, and 0.77% OA reductions respectively. The consistent degradation across all datasets indicates this component's role in preserving spectral information while maintaining sequential transformations.

Gate Mechanism (wo_gate): Removal led to 2.50%, 5.15%, and 1.77% OA decreases. Kennedy Space Center exhibited largest degradation (5.15% OA, 4.73% Kappa), indicating the gate's critical importance for adaptive feature selection in distinguishing subtle spectral differences among wetland vegetation types.

Time-Step Modulation (wo_timestep): Ablation produced 2.57%, 0.88%, and 1.41% OA reductions. While showing moderate impact individually, this component provides adaptive temporal scaling that works synergistically with other modules to enhance model flexibility.

Attention Enhancement (wo_attention): Removal caused 1.24%, 8.33%, and 2.04% OA decreases. Kennedy Space

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TABLE VIII
ABLATION STUDY

Category	Metric/Module	Full Model	wo_dual	wo_direct	wo_gate	wo_timestep	wo_attention	wo_multiscale	relu	2scales	4scales
Configuration	DST	√	×	√	√	√	√	√	√	√	√
	Direct	√	√	×	√	√	√	√	√	√	√
	Gate	√	√	√	×	√	√	√	√	√	√
	Time	√	√	√	√	×	√	√	√	√	√
	Attn	√	√	√	√	√	×	√	√	√	√
	Multi	√	√	√	√	√	√	×	√	√	√
	ReLU	×	×	×	×	×	×	×	√	×	×
	Scales	3	3	3	3	3	3	3	3	2	4
Indian Pines	OA	96.11	95.68	95.71	93.61	93.54	94.87	95.68	94.84	93.33	95.20
	AA	91.56	96.39	96.37	95.20	95.23	94.13	95.31	96.54	95.88	96.26
	Kappa	95.57	95.05	95.11	92.72	92.59	94.12	95.06	94.10	92.41	94.51
	Avg Epoch Time (s)	0.453	0.408	0.414	0.408	0.432	0.427	0.322	0.444	0.392	0.49
	Avg Inference Time (ms)	0.19	0.178	0.176	0.174	0.183	0.182	0.159	0.187	0.174	0.195
Kennedy Space Center	OA	95.34	93.75	94.17	90.19	94.46	87.01	64.54	91.34	95.75	95.77
	AA	91.29	90.03	89.37	81.62	87.50	75.99	53.86	82.24	92.35	92.36
	Kappa	94.80	93.05	93.51	89.07	93.82	85.51	60.46	90.35	95.27	95.29
	Avg Epoch Time (s)	0.249	0.24	0.242	0.242	0.243	0.242	0.22	0.245	0.234	0.257
	Avg Inference Time (ms)	0.544	0.537	0.53	0.534	0.532	0.533	0.52	0.533	0.529	0.539
Houston2018	OA	82.35	79.28	81.58	80.58	80.94	80.31	67.36	77.13	76.36	83.47
	AA	74.01	72.61	72.14	71.07	70.72	70.73	47.78	64.55	64.14	75.81
	Kappa	76.46	73.42	75.36	74.29	74.53	74.36	55.98	69.07	68.67	78.37
	Avg Epoch Time (s)	44.926	38.563	40.875	39.558	40.977	40.984	22.442	43.202	35.107	51.472
	Avg Inference Time (ms)	0.07	0.063	0.063	0.061	0.064	0.064	0.038	0.067	0.056	0.078

Center's substantial 8.33% drop (9.29% Kappa drop) demonstrates attention's crucial role in complex ecological environments, enabling focus on discriminative spectral-spatial features while suppressing redundancy.

Multiscale Branches (wo_multisc): Ablation resulted in dramatic performance degradation of 0.43%, 30.80%, and 14.99% OA across the three datasets. The severe impact on Kennedy Space Center (30.80% OA drop, 34.34% Kappa drop) and Houston2018 (14.99% OA drop, 21.48% Kappa drop) exhibits multiscale decomposition is the most critical component, essential for capturing hierarchical representations and handling complex spectral-spatial patterns in challenging datasets.

Activation Function (wo_relu): Replacing GELU with ReLU caused 1.27%, 4.00%, and 5.22% OA drops. Houston2018 and Kennedy Space Center showed substantial degradation, demonstrating GELU's smooth, non-monotonic activation profile is particularly beneficial for complex spectral characteristics in urban and wetland scenes.

Two-Scale Configuration (wo_2scales): Reducing to two parallel branches resulted in 2.78% and 5.99% OA decreases on Indian Pines and Houston2018, while surprisingly improving Kennedy Space Center performance by 0.41%. This suggests the optimal number of scales varies by dataset characteristics, with simpler architectures potentially beneficial for certain data distributions.

Four-Scale Configuration (wo_4scales): Extending to four scales showed mixed results: 0.91% decrease on Indian Pines, 2.98% decrease on Kennedy Space Center, but 1.12% improvement on Houston2018. This indicates excessive architectural complexity may lead to overfitting on smaller datasets while potentially benefiting larger, more complex urban scenes.

Computational Efficiency Analysis: Beyond classification accuracy, we evaluated computational efficiency across all ablation variants. Table VIII reports average epoch training time and inference time per image. The complete HierarchicalMamba achieves training times of 0.453, 0.249, and 44.926 s/epoch on Indian Pines, Kennedy Space Center, and Houston2018, with corresponding inference times of 0.19, 0.544, and 0.07 ms respectively. Multiscale branches constitute the primary computational component, with removal reducing training time by 29%, 12%, and 50% across datasets. However, given dramatic accuracy losses (0.43%–30.80% OA), this computational investment provesessential. Attention mechanism manifests excellent efficiency, adding only 3%–9% training overhead while providing 1.24%–8.33% accuracy gains. Lightweight components (gate, time-step modulation, direct term D) exhibit minimal runtime variations (<10% training time) while consistently improving accuracy, validating efficient design. Scale configuration analysis confirms three-scale architecture provides optimal balance: two-scale reduces training time by 13%–22% but sacrifices 2.78%–5.99% accuracy, while four-scale increases cost with mixed results. These findings demonstrate HierarchicalMamba achieves superior accuracy-efficiency trade-off through careful architectural design.

The ablation study validates both theoretical innovations and architectural efficacy of HierarchicalMamba. The DST mechanism, representing a core theoretical contribution, reflects increasing importance with dataset complexity (0.43%, 1.59%,

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and 3.07% OA degradation across Indian Pines, Kennedy Space Center, and Houston2018 respectively), particularly critical for heterogeneous urban environments where higher-order state dependencies and nonlinear dynamics dominate spectral-spatial patterns. Multiscale branches emerge as the performance-critical component, with ablation causing substantial degradation (30.80% and 14.99% OA loss on Kennedy Space Center and Houston2018), establishing their essential role in hierarchical feature extraction. Attention enhancement proves significant for complex ecological scenarios (8.33% OA loss on Kennedy Space Center), while gate mechanism (1.77%–5.15% loss) and activation function selection (1.27%–5.22% loss) contribute notably to overall performance. Time-step modulation and direct term D provide consistent moderate improvements across datasets. Component importance exhibits clear dataset-dependent patterns: simpler datasets show modest sensitivity to individual ablations, whereas challenging datasets demonstrate strong reliance on multiscale decomposition and attention mechanisms. These findings substantiate the architectural design rationale and elucidate the mechanisms underlying superior classification performance.

8) Comprehensive Performance Analysis: To provide a thorough understanding of the experimental results, this section presents a systematic analysis of the comparative performance across all methods and datasets. The analysis encompasses cross-dataset consistency patterns, quantitative stability assessments, and detailed category-level performance evaluations to comprehensively characterize the strengths and limitations of each approach.

a) Cross-dataset performance consistency analysis: Statistical analysis of all methods' performance across the five datasets reveals distinct performance patterns. All methods performed better on Pavia University and Salinas Valley datasets compared to Indian Pines, Kennedy Space Center, and Houston2018. This performance difference primarily stems from intrinsic feature complexity differences among datasets. The Pavia University dataset, as an urban scene with clearly defined land cover boundaries, enables rapid feature learning. Salinas Valley's agricultural scenes, though complex, possess relatively regular spectral characteristics. Indian Pines suffers from severe class imbalance issues. Kennedy Space Center's wetland categories exhibit high spectral similarity. Houston2018, with its high spatial resolution (0.5 m), 20 diverse urban land cover classes, severe spectral confusion among pavement and building materials, and substantial class imbalance, represents the most challenging benchmark.

b) Cross-dataset robustness evaluation: Cross-dataset robustness was quantified through performance gap analysis, defined as the difference between maximum and minimum OA across the five benchmark datasets. HierarchicalMamba exhibited a performance gap of 15.68 percentage points (ranging from 98.03% on Pavia University to 82.35% on Houston2018), demonstrating superior consistency compared to methods achieving comparable mean accuracy. Traditional convolutional architectures manifested substantial performance degradation, with gaps ranging from 27.35 to 35.62 percentage points. Attention-enhanced methods, despite achieving strong performance on conventional benchmarks, exhibited pronounced sensitivity to dataset complexity: ATN_spectral demonstrated a 48.39 percentage point gap (96.34%–47.95%). Transformer-based approaches showed moderate variability, with performance gaps ranging from 20.26 to 24.51 percentage points across architectural variants. Notably, HierarchicalMamba maintained accuracy exceeding 82% across all evaluation scenarios, whereas most comparative methods experienced substantial degradation below 75% on challenging datasets. This consistent performance across diverse geographical contexts—agricultural landscapes, urban environments, and wetland ecosystems—substantiates the robust generalization capability of the proposed unified spatial-sequential modeling framework.

c) Category-level performance in-depth analysis: Category-level analysis reveals HierarchicalMamba's significant advantages in processing difficult categories. On the small-sample Oats category in Indian Pines, HierarchicalMamba achieved 100% accuracy, with only TRANSF_pvt_tiny reaching the same level among other methods, while most achieved below 50% accuracy.

On the technically challenging Gravel category in Pavia University, HierarchicalMamba achieved 97.67% accuracy, significantly surpassing other methods (second-best being cnn_inception's 90.47%). On the complex Cabbage palm hammock category in Kennedy Space Center, HierarchicalMamba achieved 99.19% accuracy, while most methods achieved below 90%, with some below 50%.

These category-level results collectively confirm that HierarchicalMamba's theoretical innovations translate into practical advantages across the most challenging classification scenarios, validating the effectiveness of its multiscale state space modeling approach in capturing complex spectral-spatial relationships that traditional methods fail to adequately represent.

d) Architectural mechanism comparison with SSM baselines: To validate fundamental contributions beyond incremental modifications, we provide comparative analysis against three baselines. DualPathmamba employs dual-path architecture processing spectral and spatial information separately, requiring independent parameters and explicit fusion. Our unified encoding eliminates this duality through coordinate-to-index mapping, achieving +2.72%, +1.34%, and +1.77% OA improvements on Indian Pines, Pavia University, and Kennedy Space Center respectively, with particularly notable +17.51% gain on Houston2018 (82.35% versus 64.84%). HyperspectralMamba integrates SSM with adaptive recalibration through dual-stream architecture. Our SSM-internal multiscale decomposition differs fundamentally, with ablation studies (Table VIII) demonstrating substantial contribution (up to –30.80% OA when removed). Performance advantages include +9.68% OA on Indian Pines (96.11% versus 86.43%), +8.77% on Pavia University, and +6.54% on Kennedy Space Center. MSDANet achieves multiscale processing through external convolutional pyramids with attention mechanisms. Our approach extends the state space formulation itself, demonstrating +5.50%, +0.85%, +0.50%, and +16.26% OA improvements on Indian Pines, Pavia University, Salinas Val-

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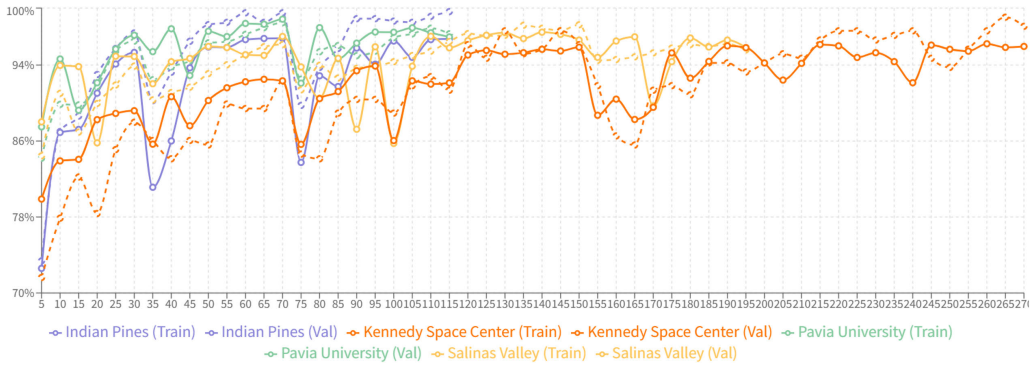


Fig. 7. Training versus validation accuracy comparison.

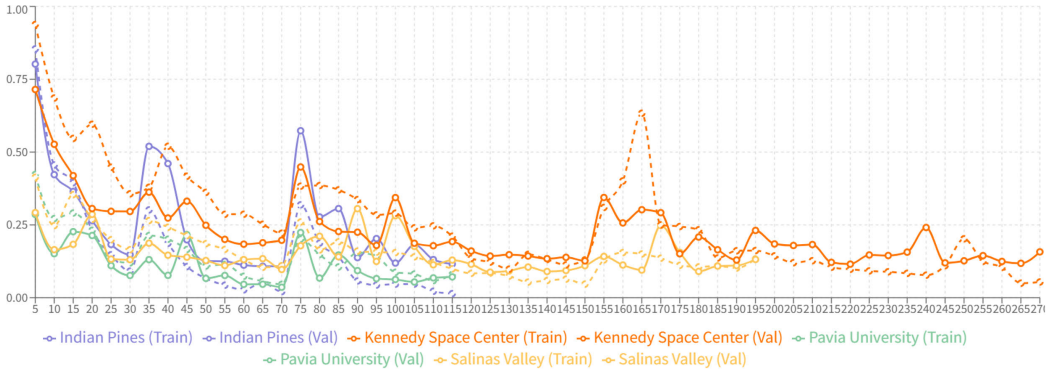


Fig. 8. Training versus validation loss comparison.

ley, and Kennedy Space Center respectively, while maintaining linear complexity $O(L)$ with 3.04M parameters. Consistency across baselines and datasets validates that unified encoding, internal multiscale decomposition, and dual transitions constitute substantial framework advancements rather than architectural variations.

9) Training Process Analysis: Based on the training data from the HierarchicalMamba HSI classification experiments and the corresponding figures, the following objective analysis is presented.

a) Distinct convergence patterns across datasets: Fig. 7 clearly illustrates different convergence characteristics among the four datasets. Pavia University (green lines) exhibits an ideal rapid convergence pattern, with validation accuracy swiftly increasing from 87% to 97% within the first 30 epochs and maintaining stability thereafter. Indian Pines (blue lines) displays oscillatory convergence behavior, with notable performance degradations at epochs 35 and 75, where validation accuracy drops from 95% to approximately 81%, indicating training instability. Kennedy Space Center (orange lines) requires the longest training cycle (270 epochs) with frequent and substantial performance fluctuations throughout the process. Salinas Valley (yellow lines) reflects a gradual convergence pattern that, while requiring 199 epochs, shows consistent upward trends.

b) Loss function behavior reflects dataset complexity: Fig. 8 reveals distinct learning difficulty hierarchies. Pavia University achieves rapid convergence of both training and validation losses to below 0.1, maintaining consistency throughout training, which indicates high class separability within the dataset. Kennedy Space Center presents the most challenging convergence behavior, with initial loss values approaching 1.0 and multiple rebound episodes during the convergence process, ultimately stabilizing at relatively higher levels. Indian Pines and Salinas Valley exhibit intermediate loss convergence patterns, though both display training-validation loss divergence, particularly in later training phases.

c) Quantitative assessment of generalization capability: Fig. 9 provides direct visualization of overfitting risks through the training-validation accuracy differential. Pavia University illustrates the smallest and most stable generalization gap, predominantly maintaining differences within $\pm 3\%$, with instances where validation performance exceeds training performance, representing ideal generalization behavior. Indian Pines exhibits significant generalization gap peaks (approximately 8%–10%) around epochs 35 and 85, corresponding to the performance degradations observed in Fig. 9 Kennedy Space Center shows volatile generalization gap fluctuations, with some convergence during epochs 150–200, though overall trends suggest overfitting tendencies.

d) Performance benchmarking on pavia university dataset: Integrating the comparative data from Table IX, HierarchicalMamba achieves performance on the Pavia University dataset using 3037 698 parameters across 119 training epochs, with an average training time of 5.87 s/epoch and inference time of 2.511 ms. Compared to traditional CNN approaches, it demonstrates superior parameter efficiency, while significantly reducing computational overhead relative to

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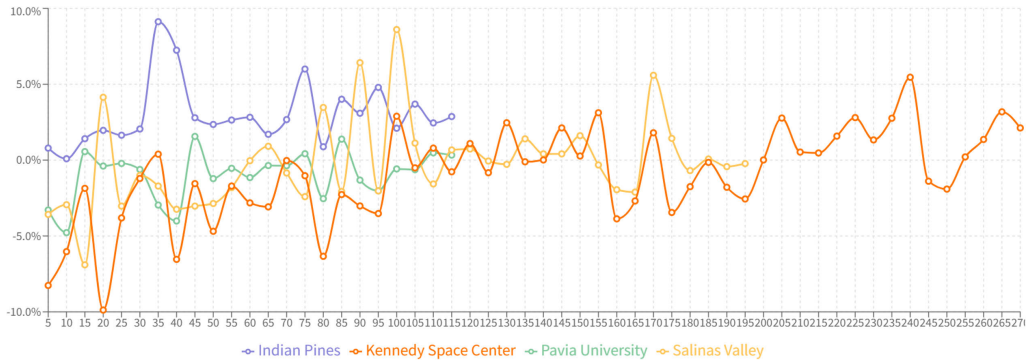


Fig. 9. Generalization gap analysis.

TABLE IX
COMPUTATIONAL PERFORMANCE METRICS ON PAVIA UNIVERSITY DATASET

Method	Total Parameters	FLOPs	Model Size (MB)	Epochs	Peak GPU Memory (MB)	Total Training Time (s)	Avg Epoch Time (s)	Avg Inference Time (ms)
cnn_deep	5,735,241	12.159M	21.88	116	105.05	236.17	2.036	0.881
cnn_inception	3,665,417	7.786M	13.98	118	73.61	295.76	2.506	1.095
cnn_efficientnetv2	4,838,609	78.620M	18.46	134	90.61	623.15	4.650	2.002
ATN_cbam	189,869	206.758K	0.72	319	19.17	538.04	1.686	0.727
ATN_nonlocal	209,899	1.251M	0.80	192	19.50	304.06	1.583	0.676
ATN_spectral	201,930	736.835K	0.77	189	19.35	279.95	1.481	0.636
ATN_dynamic	234,878	1.833M	0.90	194	19.85	339.70	1.751	0.758
TRANSF_pvt_tiny	11,793,545	633.953M	44.99	71	183.23	325.42	4.583	1.978
TRANSF_uniformer_tiny	17,067,273	53.244G	65.11	87	281.58	27259.88	313.332	129.113
TRANSF_twins_tiny	12,342,729	16.965G	47.08	183	209.48	40364.49	220.571	88.450
TRANSF_mobilevit_tiny	995,513	36.607M	3.80	143	31.57	956.48	6.688	2.889
MSDANet	4,147,694	4.148M	15.82	166	79.72	555.74	3.348	1.497
DualPathmamba	1,987,081	79.646M	7.58	196	46.58	364.86	1.861	0.839
HyperspectralMamba	11,820,553	521.025M	45.09	118	199.63	385.74	3.269	1.410
HierarchicalMamba	3,037,698	49.494M	11.59	119	63.04	698.55	5.870	2.511

Transformer-based architectures. For instance, TRANSF_uniformer_tiny requires 17 067 273 parameters and 313.332 s/epoch, whereas HierarchicalMamba achieves comparable performance with approximately one-sixth the parameters and one-fiftieth the training time.

e) Impact of dataset characteristics on model performance: The performance variations of HierarchicalMamba across the four datasets reflect their intrinsic properties: Pavia University, as an urban scene dataset with clearly defined land cover boundaries, enables rapid feature learning; Salinas Valley's agricultural scenes, though complex, possess relatively regular spectral characteristics that support stable progressive learning; Indian Pines likely suffers from severe class imbalance issues, resulting in training instability; Kennedy Space Center's complex environmental conditions and spectral confusion render it the most challenging dataset.

f) Architectural adaptability limitations: While HierarchicalMamba evinces improved performance on Pavia University, the performance variations across other datasets suggest that its architectural design may be optimized for specific types of hyperspectral data. The cross-dataset performance variations shown in Figs. 6 and 8 indicate the necessity for differentiated training strategies or architectural modifications tailored to different data characteristics.

The current unified training configuration (identical learning rate scheduling, optimizer settings, etc.) fails to adequately address the specific requirements of each dataset.

g) Computational Efficiency and Accuracy Trade-Offs: Table IX indicates that HierarchicalMamba achieves a reasonable balance in computational efficiency. Compared to lightweight ATN-series models (approximately 200 000 parameters), HierarchicalMamba sacrifices some inference speed but gains superior feature learning capabilities; compared to large-scale Transformer models, it maintains comparable accuracy while significantly reducing computational costs. This trade-off holds substantial implications for practical HSI processing applications.

h) Computational complexity and resource requirements: Table IX quantifies HierarchicalMamba's efficiency with 3.04M parameters and 49.49M FLOPs. TRANSF_pvt_tiny requires 11.79M parameters (3.88×) and 633.95M FLOPs (12.81×), while TRANSF_uniformer_tiny demands 17.07M parameters (5.62×) and 53.24G FLOPs (1076×).

Among SSM methods, DualPathmamba achieves 1.99M parameters and 79.65M FLOPs through simplified dual-path design, while HyperspectralMamba requires 11.82M parameters and 521.03M FLOPs for dual-stream architecture. HierarchicalMamba balances architectural sophistication with

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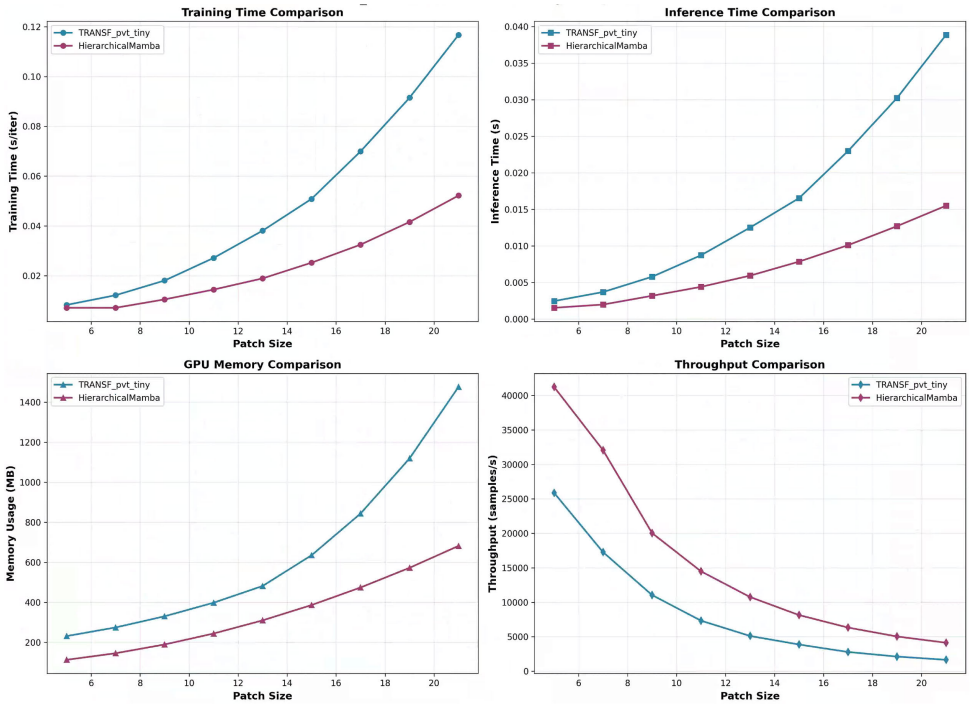


Fig. 10. Scaling complexity analysis: HierarchicalMamba versus Transformer architecture on Indian Pines dataset.

efficiency, achieving 82.35% OA on Houston2018 versus DualPathMamba's 80.17%.

Training metrics reveal distinct characteristics. HierarchicalMamba completes 119 epochs in 698.55 s (5.87 s/epoch). TRANSF_uniformer_tiny requires 27259.88 s (313.33 s/epoch, 53.4× slower). TRANSF_twins_tiny needs 220.57 s/epoch despite spatially separable attention. DualPathmamba achieves fastest training (1.86 s/epoch, 0.84 ms inference) through streamlined fusion. HyperspectralMamba requires 3.27 s/epoch and 1.41 ms inference for dual-stream coordination.

Inference performance validates deployment feasibility. HierarchicalMamba achieves 2.51 ms latency. TRANSF_uniformer_tiny requires 129.11 ms (51.4×) and TRANSF_twins_tiny needs 88.45 ms (35.2×).

Peak Memory: HierarchicalMamba 63.04 MB versus TRANSF_uniformer_tiny 281.58 MB (4.47×) and TRANSF_twins_tiny 209.48 MB (3.32×).

i) Multi-dimensional scaling validation: Systematic scaling experiments on Indian Pines dataset compared HierarchicalMamba with TRANSF_pvt_tiny under identical hardware (NVIDIA RTX 4090). Fig. 10 presents comprehensive scaling analysis across spatial dimensions (5×5–21×21 patches).

Training time indicates contrasting scaling behaviors. HierarchicalMamba exhibits near-linear growth: 8–52 ms (6.5×) for 17.64× sequence expansion ($L = H \times W$ from 25 to 441). TRANSF_pvt_tiny shows quadratic degradation: 11–117 ms (10.64×). At maximum scale, Transformer requires 2.25× longer training. Measured exponents: HierarchicalMamba ≈ 1.02 [approximating $O(L)$], TRANSF_pvt_tiny ≈ 1.73 (approaching $O(L^2)$).

Inference latency reveals similar patterns. HierarchicalMamba scales from 2 to 15 ms (7.5×), while TRANSF_pvt_tiny degrades from 2 to 38 ms (19×), yielding 2.53× speedup at maximum scale.

Memory consumption shows pronounced advantages. HierarchicalMamba increases from 104 to 680 MB (6.54×), exhibiting sub-linear scaling. TRANSF_pvt_tiny requires 222 MB to 1505 MB (6.78×), consuming 2.21× more at maximum scale. Self-attention maintains $O(L^2)$ matrices per head; SSM requires only $O(L)$ state vectors.

Processing throughput provides deployment metrics. Both architectures exceed 25 000 samples/s at 5×5 patches. At 21×21 patches, Transformer throughput degrades to 1800 samples/s, while HierarchicalMamba maintains 5200 samples/s (2.89× higher capacity). Empirical scaling aligns with theoretical analysis. Results across spatial dimensions and architectures confirm efficiency gains stem from linear SSM transitions versus quadratic attention. HierarchicalMamba's unified framework maintains $O(L)$ complexity while incorporating sophisticated spatial-spectral modeling for heterogeneous environments.

j) Cross-dataset spectral-spatial complexity analysis: To empirically validate theoretical complexity claims across spectral and spatial dimensions, computational costs were analyzed across five benchmark datasets with naturally varying configurations (see Table X). These datasets span a 4.25-fold spectral range (48–204 bands) and 272-fold spatial range (21K–5.7M pixels). Notably, Houston2018 with 48 bands requires 44.926 s training, while Indian Pines with 200 bands requires only 0.912 s, confirming independence from spectral dimensionality. The 49-fold training time variation correlates with spatial extent rather than spectral configuration, validating $O(L)$ complexity. Memory consumption remains consistent at 63–65 MB across datasets with 3–4× spectral variation, with Houston2018's elevated memory

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TABLE X
CROSS-DATASET SPECTRAL-SPATIAL COMPLEXITY ANALYSIS WITH HIERARCHICALMAMBA

Method	Bands (C)	Spatial ($H\times W$)	$L=H\times W$	Total Parameters	FLOPs	Peak GPU Memory (MB)	Avg Epoch Time (s)	Avg Inference Time (ms)
Houston2018	48	4768×1202	5,730,336	3,037,005	34.350M	135.80	44.926	0.070
Pavia University	103	610×340	207,400	3,037,698	49.494M	63.04	5.870	2.511
Kennedy Space Center	176	512×614	314,368	3,043,398	144.393M	63.13	0.934	0.370
Indian Pines	200	145×145	21,025	3,045,705	186.181M	64.85	0.912	0.332
Salinas Valley	204	512×217	111,104	3,045,961	193.668M	63.17	7.500	3.201

TABLE XI
COMPARISON WITH RECENT MAMBA-BASED METHODS FOR HSI CLASSIFICATION

Method	Indian Pines				Pavia University			
	Train/%	OA/%	AA/%	Kappa	Train/%	OA/%	AA/%	Kappa
MiM	6.7	92.08	94.11	90.96	8.9	91.58	92.76	89.83
SpecSpatMamba[76]	10.0	95.11	92.33	94.43	5.0	98.61	98.04	98.16
WaveMamba	-	-	-	-	5.0	93.11	95.79	94.43
HierarchicalMamba	8.0	96.11	91.56	95.57	4.0	98.03	97.62	97.38

TABLE XII
COMPARISON OF SPATIAL-TO-SEQUENCE SCANNING STRATEGIES (%)

Dataset	Metric	Row-major (Ours)	Bi-directional	Cross-scan 4-way
Indian Pines	OA	**96.11**	94.77	94.35
	AA	**91.56**	91.66	91.71
	Kappa	**95.57**	94.03	93.57
Pavia University	OA	**98.03**	97.53	97.21
	AA	**97.62**	96.69	96.50
	Kappa	**97.38**	96.71	96.28
Salinas Valley	OA	**97.06**	96.64	87.23
	AA	**98.80**	98.23	95.94
	Kappa	**96.73**	96.26	85.93
Kennedy Space Center	OA	**95.34**	94.07	94.69
	AA	**91.29**	88.55	91.55
	Kappa	**94.80**	93.38	94.09

Note: Row-major scanning preserves horizontal adjacency following sensor acquisition patterns, while bi-directional and cross-scan 4-way strategies employ multi-directional scanning mechanisms.

(135.80 MB) attributable to spatial activation storage for 5.7M pixels. These results align with Section II-D: spectral dimension C affects only Input Projection $O(CDL)$, accounting for less than 5% of computation, while SSM processing $O(LD)$ remains invariant to spectral bands, demonstrating that computational cost is governed by spatial sequence length rather than spectral dimensionality.

10) Analysis of Scanning Strategies: Table XII compares row-major scanning with bi-directional and cross-scan four-way across four datasets. Row-major achieves the highest average OA (96.64%), outperforming bi-directional (95.75%) and cross-scan four-way (93.37%).

Performance analysis reveals distinct patterns: On Indian Pines, row-major attains 96.11% OA, surpassing alternatives by 1.34–1.76 points with 95.57% Kappa coefficient. For Pavia University, row-major achieves 98.03% OA while maintaining lower computational cost than cross-scan four-way (97.21% OA, 4× overhead). The most substantial advantage emerges on Salinas Valley (97.06% versus 87.23% for cross-scan four-way, +9.83 points), indicating that multi-directional scanning introduces detrimental feature redundancy for agricultural vegetation classification. On Kennedy Space Center, row-major obtains 95.34% OA and 91.29% AA, exceeding bi-directional by 1.27 and 2.74 points respectively, demonstrating superior per-class discrimination for wetland ecosystems.

These findings validate that row-major scanning provides optimal accuracy for HSI classification, naturally aligning with horizontal sensor acquisition patterns in remote sensing systems.

This validates the spatial proximity preservation mechanism described in Section II-A1: row-major ordering maintains deterministic distance patterns enabling efficient spatial dependency learning, whereas multi-directional strategies disrupt structural coherence, introducing detrimental redundancy evident in Salinas Valley's substantial performance degradation.

11) Internal Mechanism Visualization: To elucidate the effectiveness of our architectural innovations, we present visualizations of internal mechanisms extracted from the Indian Pines dataset in Figs. 11 and 12. Fig. 11(a)–(c) illustrate the DST process through three complementary perspectives. The first transition [see Fig. 11(a)] establishes initial feature representations through matrix multiplication with A, while the second transition [see Fig. 11(b)] refines these representations after gating and projection operations. The state evolution

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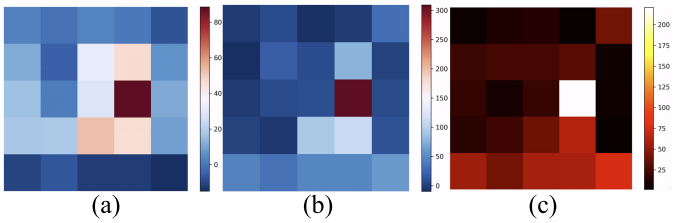


Fig. 11. DST. (a) First state transition. (b) Second state transition state. (c) State evolution.

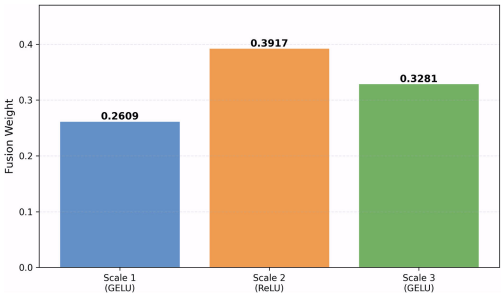


Fig. 12. Adaptive fusion weights.

map [see Fig. 11(c)] quantifies the magnitude of changes between successive transitions, revealing substantial variations spanning 0–200+ units across spatial locations. These observations substantiate that applying the state transformation twice enables the model to capture higher-order dependencies that would remain inaccessible through conventional single-stage processing.

Fig. 12 depicts learned fusion weights and branch-specific feature characteristics. The model adaptively allocates weights—scale 2: 0.3917, scale 3: 0.3281, scale 1: 0.2609—contrasting with fixed averaging schemes. Feature distribution analysis on Indian Pines dataset reveals functional specialization: GELU-activated branches produce continuous, approximately Gaussian distributions modeling gradual spectral transitions, whereas ReLU activation generates positively-skewed sparse distributions emphasizing discriminative boundaries between crop categories.

12) Comparison With Recent Mamba-Based Methods: Table VI presents comprehensive comparisons with four recent Mamba-based architectures specifically designed for HSI classification: MiM [X], SpecSpatMamba [Y], WaveMamba [Z], and DualPathMamba [60]. Experimental results demonstrate that HierarchicalMamba achieves consistently superior or competitive performance across both datasets.

On Indian Pines, HierarchicalMamba attains 96.11% OA, 91.56% AA, and 95.57% Kappa with 8.0% training samples, equaling DualPathMamba while substantially outperforming MiM (92.08% OA, 94.11% AA) and SpecSpatMamba (95.11% OA, 92.33% AA). Notably, our method achieves this performance with a training ratio comparable to other approaches, demonstrating efficient learning capability. On Pavia University, HierarchicalMamba achieves 98.03% OA, 97.62% AA, and 97.38% Kappa using merely 4.0% training samples, demonstrating performance comparable to SpecSpatMamba (98.61% OA with 5.0% training samples) and significantly exceeding WaveMamba (93.11% OA with 5.0% training samples). The slight advantage of SpecSpatMamba comes at the cost of 25% more training data, indicating HierarchicalMamba's superior sample efficiency.

These results validate the effectiveness of HierarchicalMamba's DST mechanism and hierarchical multiscale architecture in capturing complex spectral-spatial dependencies, positioning it among the leading Mamba-based approaches for HSI classification.

13) HierarchicalMamba Theoretical Advantages and Innovation Analysis: The experimental results demonstrate that HierarchicalMamba's superior performance can be attributed to its core theoretical innovations.

1) *Contribution of Unified Spatial-Sequential Modeling Framework:* The experimental results validated the effectiveness of the spatial-sequential transformation mechanism proposed in this work. Through encoding 2-D spatial coordinate information into positional indices of 1-D sequences, HierarchicalMamba achieves unified modeling of spectral correlations and spatial neighborhood relationships, avoiding the design complexity and computational redundancy of traditional dual-stream network architectures. This approach showed consistent performance across all five datasets, with particularly notable results on the spatially complex Pavia University dataset.
2) *Multiscale SSM Decomposition Theory:* The algorithm's heterogeneous parallel branch design implements multiscale feature decomposition, utilizing GELU activation functions to capture gradual features in spectral data while employing ReLU activation functions to extract sparse discriminative features. Combined with adaptive learnable fusion weight mechanisms, this enables feature combination at different granularities. The experimental results indicated this approach's adaptability when processing datasets with different spectral complexities and category distribution characteristics, maintaining classification performance across diverse environments including the complex wetland ecosystem of the Kennedy Space Center dataset.
3) *DST Mechanism:* Through DST matrix operations within each computational cycle, HierarchicalMamba enables modeling of higher-order state dependencies through dual matrix applications. The experimental data indicated that this mechanism enhanced the model's sensitivity for subtle inter-class spectral differences, showing performance advantages in small-sample category and similar category discrimination tasks across datasets.

The experimental evidence supported its effectiveness across multiple benchmark datasets, contributing to both SSM methodology and practical HSI processing applications.

IV. Conclusion

This article addresses fundamental challenges in HSI classification through HierarchicalMamba, a SSM-based

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architecture achieving linear computational complexity $O(L)$ while capturing complex spatial-spectral dependencies.

Three principal contributions are established: First, the unified spatial-sequential framework eliminates dual-stream architectural separation by encoding 2-D spatial coordinates into 1-D sequence indices, enabling simultaneous representation of spectral correlations and spatial neighborhood relationships within a single SSM pathway. Second, the multiscale SSM decomposition theory embeds multi-granularity feature extraction directly within state space formulations through heterogeneous parallel branches (GELU for smooth gradients, ReLU for sparse features) with adaptive Softmax-normalized fusion. Third, the DST mechanism applies the state matrix twice per cycle, extending first-order Markovian dynamics to capture higher-order dependencies analogous to multistep numerical integration.

Comprehensive experiments on five benchmark datasets demonstrate effectiveness: 96.11% (Indian Pines), 98.03% (Pavia University), 97.06% (Salinas Valley), 95.34% (Kennedy Space Center), and 82.35% (Houston2018) OA. Comparative analysis reveals consistent performance advantages over state-of-the-art methods including recent Mamba-based architectures. The algorithm exhibits computational efficiency with 3.04M parameters, achieving 53× faster training than representative Transformer baselines while requiring 3.88–5.62× fewer parameters.

Current limitations include fixed spatial scanning strategies, computational overhead of dual transitions, and sensitivity to fusion weight initialization. Future research directions encompass adaptive scanning mechanisms, efficient state transition optimization, multi-temporal hyperspectral extension, and uncertainty quantification integration to enhance robustness for practical deployment.

Data Availability Statement

The authors confirm that the data supporting the findings of this study are available within the article. All HSI scenes utilized in this study are publicly available on the website: Hyperspectral Remote Sensing Scenes—Grupo de Inteligencia Computacional (GIC) (ehu.eus). The source code for HierarchicalMamba will be made publicly available at: https://github.com/Jason20155/HierarchicalMamba

Competing Interests

The authors declare no competing interests.

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